

Ammonia fired gas turbines: Recent advances and future perspectives

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ABSTRACT

Transition to carbon-neutral energy generation is a reality, and great efforts are being made in this direction in the last years. Ammonia is a promising carbon-free fuel, and it can be used as a gas turbine fuel in existing power generation cycles without significant modification. This paper considers various aspects of the use of ammonia as a gas turbine fuel: it provides a review of existing ammonia-fired gas turbines and conducts a study of the prospective technology of using ammonia via thermochemical transformation into hydrogen-rich gas. Moreover, this paper considers on-board hydrogen production technology via thermochemical ammonia transformation. The ammonia fired chemically recuperated gas turbine is analyzed for which the thermal efficiency can be increased up to 5%–7% comparing to traditional gas turbines. Moreover, hydrogen-rich fuel with a hydrogen mole fraction up to 75% is used as a fuel, leading to more stable combustion with lower NO_x emission up to 6–10 ppm. Additionally, an approach to on-board hydrogen production from ammonia via the utilization of solar energy is investigated. It is shown that solar energy can replace up to 25% of heat obtained via ammonia combustion. The paper discusses future perspectives in investigations for ammonia-fired gas turbines with on-board ammonia transformation.

1. Introduction

Transition to zero carbon energy and industry is crucial for mitigating the impacts of climate change and reducing greenhouse gas emissions [1–3]. It involves a shift towards renewable energy sources such as wind, solar, and hydropower, as well as implementing energy-efficient technologies, utilization of carbon-free fuels in industries. China is the largest producer of renewable energy in the world and has set a target to reach peak carbon emissions by 2030 and achieve carbon neutrality by 2060 [4,5]. Moreover, the governments of many countries have implemented various programs to support renewable energy [6].

However, despite the progress made in the use of renewable energy sources such as solar and wind, their share in the energy balance remains relatively small [7]. The production of electricity through thermal power plants by burning fossil fuels is still dominant. Ammonia has the potential to be a way to zero carbon energy because it can be produced from renewable sources such as wind and solar power [8–10]. In addition, ammonia can be used as a fuel for power generation [11]. It has a high energy density, which makes it a viable alternative to fossil fuels such as natural gas and diesel. Additionally, ammonia does not produce greenhouse gas emissions when burned, making it a zero carbon fuel. Another advantage of ammonia is that it can be

easily transported and stored in comparing with hydrogen that can be considered as zero-carbon fuel [12]. Ammonia can be liquefied at low pressure and stored in tanks, which makes it a practical option for use in remote areas or for transportation over long distances.

Recently, ammonia has found application as a fuel for internal combustion engines [13,14], which convert chemical energy into mechanical energy. Ammonia has been recognized as a promising fuel for internal combustion engines due to its renewability, abundance, and low carbon content. There are several examples of ammonia being used as a fuel for internal combustion engines [9]. The performance of ammonia as an ICE fuel depends on several factors, including the engine design, the combustion process, and the properties of the fuel [15]. While ammonia has a lower energy content than gasoline or diesel, it has a higher octane rating, which means it can withstand higher compression ratios without causing engine knock. One of the main challenges in using ammonia as an ICE fuel is its low ignition temperature, which can be overcome by using a spark plug or preheating system. A study by Li et al. [16] found that ammonia had a higher octane rating than gasoline and diesel, making it suitable for use in high-compression engines. The study also found that ammonia had lower emissions of nitrogen oxides and particulate matter than diesel. Zhan et al. [17]

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evaluated the performance of an ammonia-fueled engine and found that it had similar power output and fuel consumption to a diesel engine. The study also found that the ammonia engine had lower emissions of carbon monoxide, nitrogen oxides, and particulate matter than the diesel engine. However, there are still challenges to overcome, such as the low flame speed and low ignition temperature of ammonia. Despite these challenges, the use of ammonia as a fuel for internal combustion engines holds promise for a more sustainable and environmentally friendly future [18]. In addition to internal combustion engines for land vehicles, ammonia is also being considered as a fuel for marine engines [19,20]. The International Maritime Organization (IMO) has set ambitious goals to reduce greenhouse gas emissions from the shipping industry by at least 50% by 2050 compared to 2008 levels [21]. One option being explored is the use of ammonia as a marine fuel. A study by Sun et al. [22] found that an ammonia-fueled engine had similar performance to a diesel engine and emitted lower levels of nitrogen oxides and particulate matter. Another study by Ejder et al. [23] found that an ammonia-fueled ship had lower greenhouse gas emissions than a conventional ship running on heavy fuel oil. Nonetheless, the potential benefits of using ammonia as a marine fuel, such as its renewability and low carbon content, make it an attractive option for meeting the CO₂ emissions reduction goals.

The use of ammonia as a fuel is promising not only in internal combustion engines and marine applications, but also in gas turbines in order to reduce carbon dioxide emissions. The use of ammonia as a fuel for gas turbines is still in the early stages of development, but there has been some promising progress in recent years. Despite researcher's and engineer's efforts, there are still some challenges that need to be addressed in order to make ammonia-fired gas turbines a viable option for commercial use. One of the main challenges is the development of effective schematic diagrams, reducing the emissions of nitrogen oxides (NO_x), which can be produced when ammonia is burned, and low combustion efficiency due to the low flame speed and long quenching distance of the fuel. This paper aims to review recent progress in ammonia-fired gas turbines, identify challenges, and propose engineering solutions for future development.

2. Gas turbines fired with pure NH₃

Gas turbines are widely used power plants. Their power ranges from a few kilowatts to hundreds of megawatts. Most existing gas turbine plants use natural gas as fuel [24]. A distinctive feature of the use of ammonia as a fuel to get a carbon-free energy is the possibility of using existing gas turbine plants after minor modifications. In general, a schematic diagram of an ammonia gas turbine consists of the following main elements: a compressor, a combustion chamber, and a gas turbine. However, when the ammonia pressure is greater than 10 bar at a temperature of about +20 °C, the ammonia is in a liquid state. Therefore, for regasification, a regasifier heat exchanger is used, in which hot water is a heat source.

In 2020, Mitsubishi Power announced the development of a 40-MW gas turbine that can directly combust 100% ammonia [25]. This 40-MW gas turbine will be based on their H-25 series. Mitsubishi's engineers and researchers suggested to combine a gas turbine with a steam power cycle to achieve high efficiency. The distinguished feature of this ammonia-fired gas turbine is a much larger combustion chamber in comparison to a combustor of natural gas fired version of this gas turbine because NH₃ fuel has a low combustion speed than natural gas. To get gaseous ammonia from the storage tank, the hot water is used as heat source. Therefore, the schematic diagram of ammonia-fired gas turbine (H-25 series) with heat recovery steam generator and steam turbine cycle can be illustrated as in Fig. 1.

There are a few existing ammonia-fired gas turbines in operation around the world. Below is some information on a few of them:

- **Turboden 15 MW ammonia-fired gas turbine:** This turbine was installed in a cogeneration plant in Germany in 2018. It has a capacity of 15 MW and is designed to operate on pure ammonia or a mixture of ammonia and natural gas. The turbine was manufactured by Turboden, an Italian company that specializes in organic Rankine cycle (ORC) and gas turbine technologies [27];
- **MAN Energy Solutions ammonia-fired gas turbine [28]:** MAN Energy Solutions is currently developing an ammonia-fired gas turbine that is expected to have a capacity of around 20 MW. The turbine is being designed to operate on pure ammonia or a mixture of ammonia and natural gas. MAN Energy Solutions is a German company that specializes in power generation and marine propulsion systems;
- **Kawasaki Heavy Industries ammonia-fired gas turbine [29]:** Kawasaki Heavy Industries has developed a 1 MW ammonia-fired gas turbine that is currently being tested in Japan. The turbine is designed to operate on pure ammonia or a mixture of ammonia and natural gas. Kawasaki Heavy Industries is a Japanese company that produces a wide range of industrial equipment, including gas turbines;
- **Siemens ammonia-fired gas turbine [30,31]:** Siemens has developed an ammonia-fired gas turbine that is currently being tested in Sweden. The turbine has a capacity of 2 MW and is designed to operate on pure ammonia or a mixture of ammonia and natural gas. Siemens is a German company that produces a wide range of industrial equipment, including gas turbines.

In terms of performance and parameters, the exact specifications of these turbines vary depending on the manufacturer and the specific model. However, all of these turbines are designed to operate on ammonia, which is a promising alternative fuel for power generation due to its zero carbon emissions. Ammonia-fired gas turbines are still relatively new technology, so there is limited data on their performance and efficiency compared to traditional gas turbines. However, early results from testing and pilot projects have been promising, and many experts believe that ammonia could play a significant role in the transition to a low-carbon energy system.

In 2023, General Electric Gas Power in a cooperation with IHI Corporation announced their plans to develop the heavy duty gas turbines [32–34]. The roadmap will include the application of GE's 6F.03, 7F and 9F gas turbines to use up to 100% ammonia safely and in a commercially competitive way. It may also be implemented in other gas turbines in the future.

Despite the prospects for the use of ammonia, there are not many real examples of the use of gas turbines. There could be several reasons for the limited information about the application of ammonia-fired gas turbines. One reason could be that this technology is still in its early stages of development and commercialization, and there may not be many companies or organizations that have implemented it yet. Another reason could be that ammonia as a fuel source is not as widely known or accepted as other fuels such as natural gas or diesel. Additionally, the use of ammonia as a fuel source may require additional safety measures and regulations, which could also limit its adoption.

Despite the limited information on the existing ammonia fired gas turbines, early researches on the development of ammonia fired gas turbines was published in the late 1960's [35,36]. These investigations revealed that ammonia had a higher ignition energy than hydrocarbon fuels because NH₃ has low reactivity. For stoichiometric conditions, the quenching distance of NH₃-air mixture is 3.5 times higher than for propane, and NH₃ can be burned at narrower equivalence ratios [36]. However, the authors discovered that NH₃ dissociation can generate faster flames, which can attain similar properties to the fossil fuels. It was found that 28% dissociated ammonia can be utilized as a substitute fuel in gas turbines designed for fossil (natural gas, propane, etc.) fuels. The performed tests showed that the main issue with using NH₃

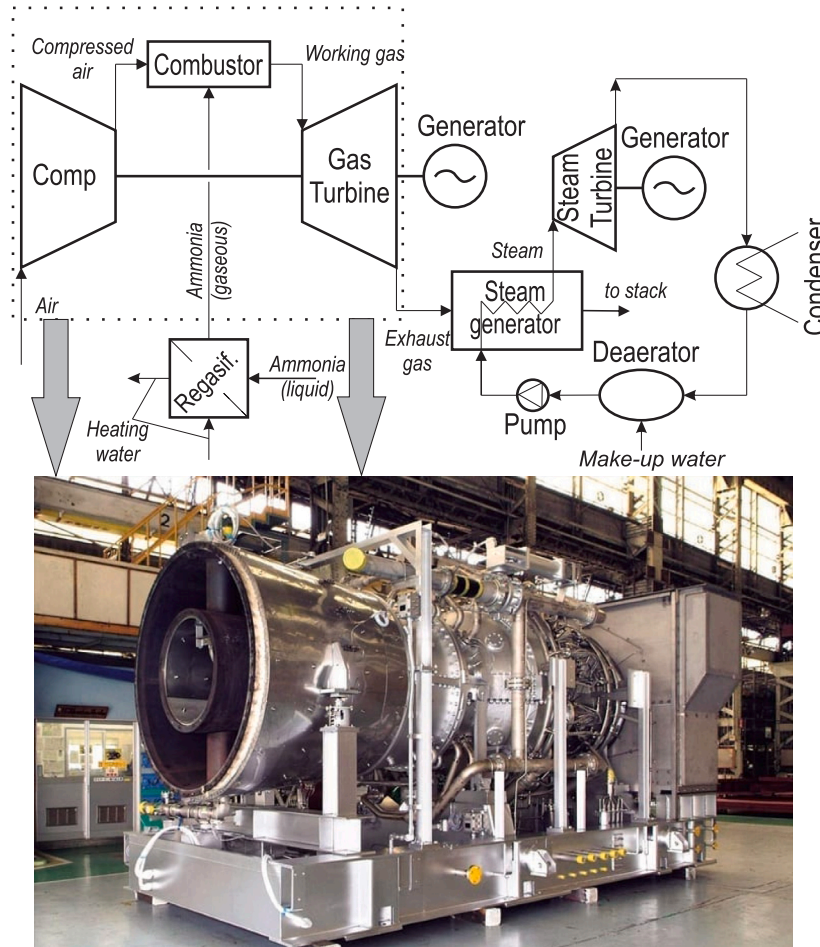


Fig. 1. Schematic diagram of ammonia-fired gas turbine (H-25 series) with steam turbine cycle from Mitsubishi Heavy Industry and Mitsubishi Power's H-25 Series gas turbines [26].

as a gas turbine fuel was the relatively slow rate of the combustion process, resulting in a laminar burning velocity of 0.06–0.08 m/s [37]. When air flow is decreased to get sufficient residence time for the combustion process, a decrease in Reynolds number effects leads a decrease in a turbulence and less effective mixing. These effects is decreasing combustion efficiency in gas turbines [38].

Moreover, in the United States in the 1960's Solar and UC Berkeley conducted research on a 250HP T-350 gas turbine [39]. The study found that ammonia had a higher ignition energy than fossil fuels, but dissociation of ammonia could produce faster flames similar to some hydrocarbons. The main issue with using ammonia-air as a turbine fuel was the relatively slow chemical reaction rate, resulting in reduced combustion efficiency. While successful power production was achieved when using ammonia as an engine fuel, it fell short in terms of emissions performance for NO_x. Overall, the study highlights the potential of ammonia as a substitute fuel in gas turbine combustion systems sized for hydrocarbon fuels.

During the 1990s and 2000s, researchers primarily focused on the combined use of ammonia and hydrocarbon fuels. This was because ammonia-hydrocarbon blends have similar combustion characteristics to traditional hydrocarbon fuels. Researchers paid significant attention to determining the emission of nitrogen oxides. In the late 1990's, Italian power generator ENEL has conducted a research program to explore the use of ammonia as a fuel for power generation [40]. Although ammonia has shown potential as a substitute fuel for gas turbine combustion systems, the study found that there are still challenges to overcome, particularly in terms of combustion efficiency and emissions

performance, in particular in NO_x emission. Nonetheless, this research is a significant step towards developing cleaner and more sustainable energy solutions for the future.

Researchers from Japan at the National Institute of Advanced Industrial Science and Technology have developed a micro gas turbine with ultra-low NO_x emissions [41–43]. The researchers focused on reducing NO_x and N₂O emissions of micro gas turbines by testing different configurations of the combustion chamber under a wide range of operating regimes, from 10 to 50 kWe. The results showed that the NO_x emissions for an ammonia-fueled gas turbine were about 337 ppm. Additionally, the authors concluded that the concentration of remaining ammonia in the flue gas was significantly reduced at low power output. They also analyzed N₂O emissions and stated that the development technology significantly reduces this pollutant.

In the 2010s, most research on the use of ammonia as a gas turbine fuel focused on the use of ammonia blends with various fuels such as methane [44–46], propane [47,48], hydrogen [49–51], etc. A set of papers was published to explore the effect of ammonia addition to the fuel on emissions and efficiency performance.

3. Gas turbines fired with ammonia blends

3.1. Combustion performances

The use of pure ammonia as a gas turbine fuel is still in the research and development stage, despite some existing applications. However, the use of ammonia as a gas turbine fuel in a mixture

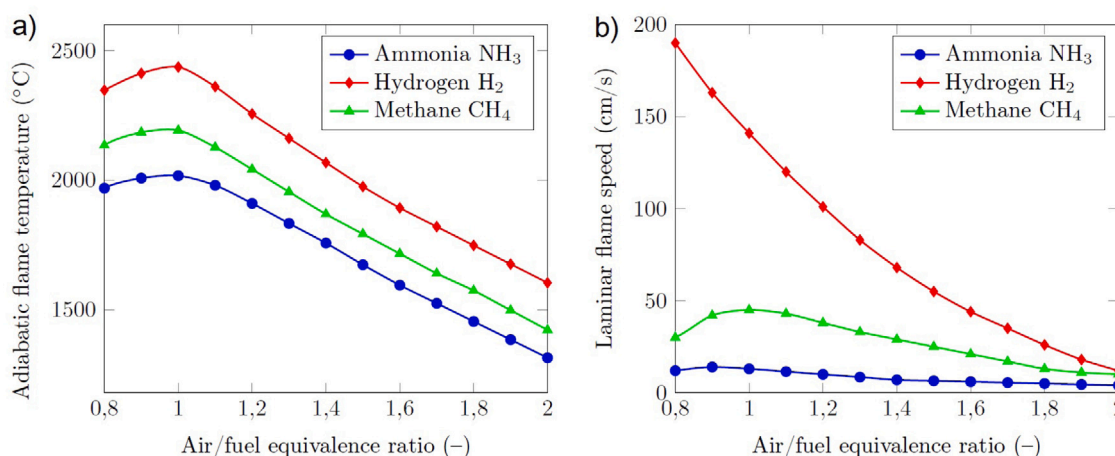


Fig. 2. A comparison of combustion performances of ammonia, hydrogen, and methane as a function of air/fuel equivalence ratio: (a) adiabatic flame temperature; (b) laminar flame speed.

with other fuels has recently piqued the interest of scientists and engineers. The benefits of using a mixture of ammonia and other fuels as a fuel in gas turbines include improved combustion efficiency and reduced emissions of pollutants such as NO_x. Ammonia has a lower heating value than conventional fuels like natural gas, so mixing it with other fuels can help to increase the overall energy density of the fuel. Additionally, blending ammonia with other fuels can help to mitigate some of the challenges associated with using pure ammonia as a fuel, such as issues with storage and handling. Overall, using a mixture of ammonia and other fuels as a fuel in gas turbines could offer a more sustainable and environmentally friendly alternative to traditional fossil fuels. Moreover, the mixing of ammonia with other fuels can lead to changes in combustion performance, such as adiabatic temperature and laminar flame speed. The most common fuels used for mixing with ammonia are methane and hydrogen. Fig. 2 illustrates a comparison of the combustion performances of ammonia, hydrogen, and methane.

Fig. 2a shows that the adiabatic temperature can be controlled by the air-fuel equivalence ratio. The actual temperature of combustion products for gas turbines is limited to 1500–1600 °C, which is only observed in high-efficiency and high-temperature gas turbines. Most existing gas turbines, especially low-power ones, operate under lower temperature regimes. Therefore, the moderately low combustion heat of ammonia can be compensated for by regulating the air-fuel equivalence ratio in a combustion chamber.

However, the situation is completely different for the laminar flame speed. Fig. 2b shows that the flame speed of ammonia at λ = 1 is about 5 times less than that for methane and about 16 times less than that for hydrogen. These proportions are observed for air-fuel equivalence ratios up to 2. Therefore, the low flame speed is one of the main limiting factors for the transition of existing gas turbines from natural gas to ammonia. The low flame speed of ammonia leads to an increase in the size of the combustion chamber of a gas turbine. Adding hydrogen or other fuels such as methane or natural gas is one way to solve this issue.

To better understand the effect of adding hydrogen or methane to ammonia, the laminar flame speed for the different blends was analyzed based on a literature review. Fig. 3 shows the laminar flame speed as a function of ammonia mole fraction for ammonia–hydrogen (a) and ammonia–methane (b) blends.

Fig. 3 demonstrates that adding hydrogen and methane to ammonia has different effects. While adding methane to ammonia results in a steady increase in laminar flame speed, adding hydrogen to ammonia produces different results due to the different densities and combustion heats of the components. The density of ammonia (under normal conditions) is 0.73 kg/m³, while the density of methane and hydrogen

is 0.657 kg/m³ and 0.09 kg/m³, respectively. Consequently, adding a small amount of hydrogen has an insignificant effect on the laminar flame speed. However, adding more than 50% hydrogen leads to a significant increase in the laminar flame speed due to the high flame speed of hydrogen, making the mass fraction of hydrogen notable, and resulting in a rapid increase in the laminar flame speed of the entire mixture.

Fig. 4 shows the dependence of ammonia mass fraction and lower heating value as a function of ammonia mole fraction for ammonia–methane and ammonia–hydrogen blends. Fig. 4a shows that adding methane to ammonia leads to a linear change in the ammonia mass fraction and lower heating value of the blend. The main reason for this is the similar densities of ammonia and methane. However, Fig. 4b depicts a different dependence when adding hydrogen to ammonia. 20%_{mol} of hydrogen gives only 2.87%_{mass}. For the ammonia–hydrogen blend with a molar composition of 1:1, the mass fraction of ammonia and hydrogen is 89.42% and 10.58%, respectively. Therefore, adding hydrogen to ammonia has a low effect when the hydrogen mole fraction is less than 0.5.

Another important factor to consider is the control of NO_x emissions, especially in high-temperature gas turbines. There is ample information in the literature that suggests NO_x can be controlled by improving the combustion process and implementing NO_x removal techniques. However, there are varying results regarding this issue, which will be discussed further.

3.2. Ammonia–hydrogen blends

The gas turbines fired with ammonia blends have found significant application in recent years because they can be considered an intermediate stage towards the use of pure ammonia combustion processes. One of the main reasons for using ammonia blends is to address the following issues associated with pure ammonia combustion:

- low flame speed and low flame temperature;
- unstable combustion and low kinetic;
- high NO_x emission.

There are numerous papers on the use of ammonia blends that focus on various aspects of gas turbines fired with ammonia blends.

The transition to carbon-zero energy is focused on the use of carbon-neutral fuels, with ammonia and hydrogen being highlighted due to their moderate widespread use and moderate cost. Research studies on ammonia–hydrogen blends for gas turbines can be divided into subgroups such as chemical kinetics mechanism, numerical simulation,

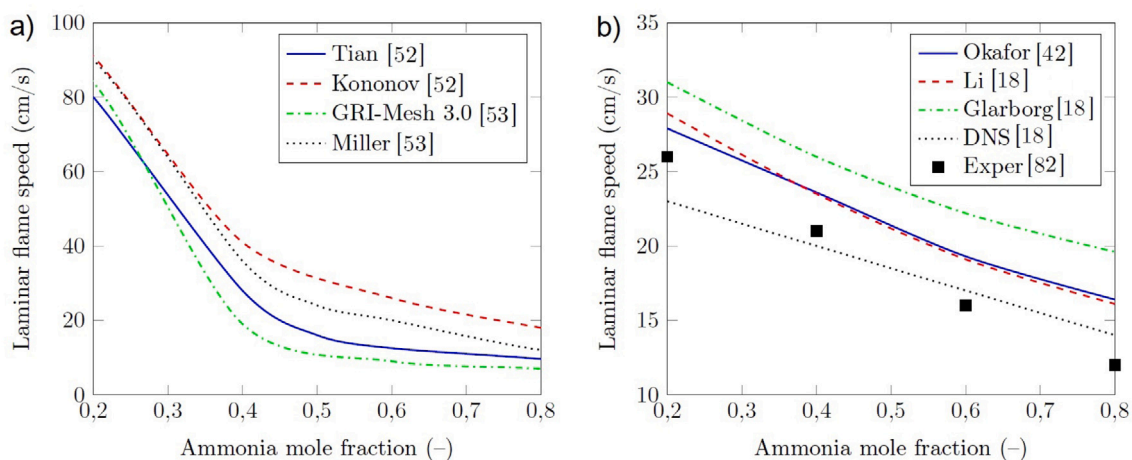


Fig. 3. Laminar flame speed for various blends: (a) ammonia-hydrogen at $\lambda = 1.0$, $p = 3$ bar; (b) ammonia-methane at $\lambda = 1.2$, $p = 3$ bar.

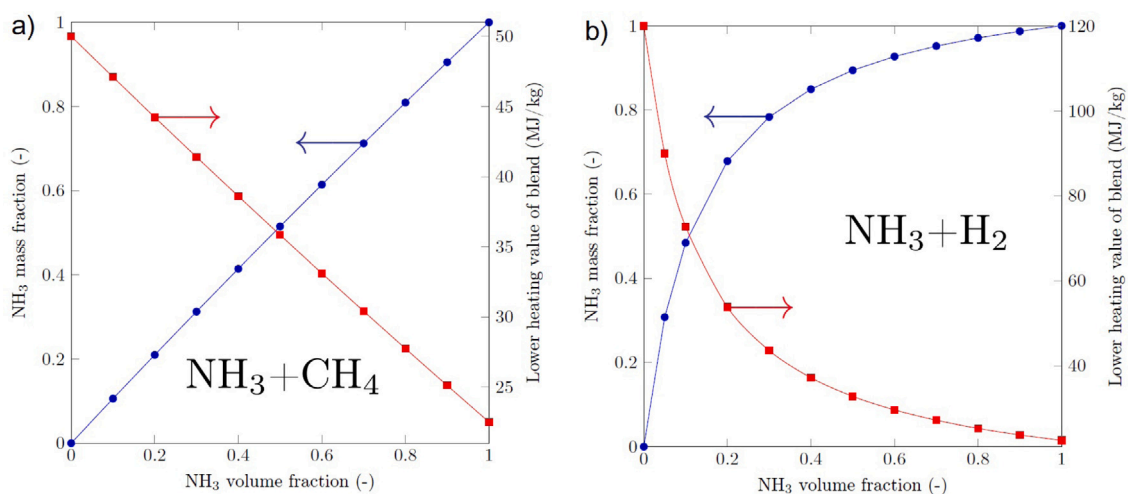


Fig. 4. Ammonia mass fraction and lower heating value of the blends as a function of ammonia mole fraction: (a) ammonia-methane ($\text{NH}_3 + \text{CH}_4$) blend; (b) ammonia-hydrogen blend ($\text{NH}_3 + \text{H}_2$).

experimental, and laboratory tests. Kumar and Meyer presented the results of an experimental and modeling study of chemical-kinetics mechanisms for hydrogen-ammonia-air mixtures [52]. They analyzed the numerical results obtained via GRI-Mech3.0, and Tian and Kononov detailed chemical kinetic mechanisms. Numerical simulation was carried out using CHEMKIN software. The most accurate results for the equivalence air-fuel ratio above 1.0 were obtained for Tian's model. The authors showed that for various compositions of ammonia-hydrogen blends, the flame speed can be the same as for methane-air combustion. The main conclusion from Kumar's paper is that adding hydrogen to ammonia makes it possible to achieve a flame speed similar to methane-air combustion. Thus, ammonia-hydrogen blends can be used for existing gas turbines without significant modification of combustion chambers.

The prediction of NO_x emission when ammonia-hydrogen blends are utilized as a gas turbine fuel can be conducted both experimentally and numerically. Xiao and Valera-Medina provided numerical analysis with experimental verification of kinetic mechanisms for $\text{NH}_3\text{-H}_2$ -air combustion for the gas turbine conditions [53]. Their paper mainly focused on investigation of flame speed and NO_x emission. In a part of flame speed, the authors obtained the results similar to the results obtained by other authors [54,55]. They established that hydrogen addition to ammonia leads to an increase in ammonia flame speed. Higher flame speed makes it possible to use ammonia-hydrogen blends in the existing gas turbines without significant modernization.

Table 1

N_2O and NO emission for various composition of ammonia-hydrogen blends [53,56] for the different equivalence air-fuel ratios at constant pressure.

NH_3	H_2	ER	N_2O (%mol)			NO (%mol)		
			Exp. [56]	GRI3.0	Miller	Exp. [56]	GRI3.0	Miller
0.7742	0.2258	1.00	0.98	0.98	0.32	1.71	2.51	1.25
0.6875	0.3125	1.00	1.01	0.96	0.33	1.62	2.36	1.18
0.6176	0.3824	1.00	1.12	0.94	0.31	1.52	2.18	1.09
0.6875	0.3125	0.90	1.38	0.78	0.38	2.78	3.19	1.63
0.6875	0.3125	1.10	1.07	0.93	0.34	1.31	2.11	1.08

Additionally, the authors analyzed NO , N_2O and NO_x emissions for various compositions of ammonia-hydrogen blends. Table 1 shows the effect of ammonia-hydrogen blend composition on N_2O and NO emissions obtained via experimental tests [56] and numerical simulation [53]. The results show that an increase in hydrogen mole fraction leads to an increase in N_2O emission and decrease in NO emission. Moreover, Table 1 presents that an increase in equivalence air-fuel ratio gives a decrease in both N_2O and NO emission. It should be noted that gas turbines are operating under equivalence air-fuel ratio notably higher than 1. Therefore, hydrogen addition to ammonia will lead to a decrease in N_2O and NO emissions.

Gas turbines operate at high pressure, with the turbine inlet pressure capable of reaching a few dozen bars. General Electric has announced

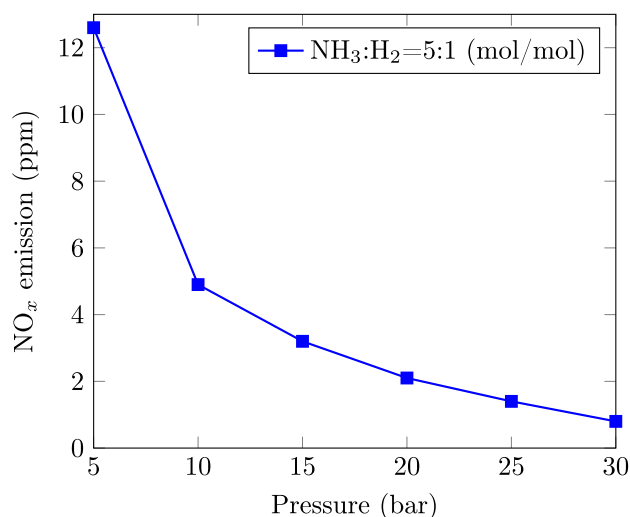


Fig. 5. Effect of pressure on NO_x emission for combustion NH₃-H₂ blend at 5:1 mol/mol ratio.

plans to modify gas turbines of the 6F, 7F, and 9F-series for the use of ammonia as a fuel [57,58]. The existing gas turbines of the 6F, 7F, and 9F-series, fired with natural gas, have NO_x emissions of about 15 ppm. Xiao and Valera-Medina investigated the effect of pressure on NO_x emissions for gas turbines fueled with ammonia-hydrogen blends [53]. Fig. 5 shows the effect of pressure on NO_x emissions for the combustion of NH₃-H₂ blend at a 5:1 mol/mol ratio. An oxygen-argon mixture was used as an oxidizer for the ammonia-hydrogen blend. Fig. 5 depicts the significant decrease in NO_x emissions when the pressure is increased. This effect can be explained by Le Chatelier's principle because the products of NO_x formation reaction contain fewer moles than the initial components. Moreover, Fig. 5 makes it clear that for pressures above 20 bar, NO_x emissions from nitrogen contained in ammonia give only a few ppm. These results illustrate that hydrogen addition to ammonia in gas turbines can be considered a promising way to reduce NO_x emissions in existing gas turbines.

Valera-Medina et al. [59] conducted numerical and experimental investigations on a premixed NH₃/H₂ combustor for gas turbines. They studied the combustion of ammonia-hydrogen blends in the presence of air and performed a set of tests on a gas turbine with a power output of approximately 40 kW. The experimental results indicated that the combustion process of a blend with NH₃ = 60%_{vol} and H₂ = 40%_{vol} is unstable, while the combustion of an ammonia-hydrogen blend with 70%_{vol} and 30%_{vol}, respectively, is stable. Additionally, the authors tested NO_x emissions for different combustor configurations and demonstrated that NO_x emissions for an air/fuel equivalence ratio above 1.3 are below 10 ppm for dry composition. NO_x close to zero is observed at an air/fuel equivalence ratio above 1.4.

Moreover, Valera-Medina et al. numerically and experimentally investigated premixed NH₃/H₂ combustor for gas turbines [59]. They investigated the combustion of ammonia-hydrogen blends in a presence of air. A set of tests was performed in gas turbine at a power output about 40 kW. The experimental indicated that the combustion process of blend at NH₃ = 60%_{vol} and H₂ = 40%_{vol} is unstable. However, combustion is stable for ammonia-hydrogen blend of 70%_{vol} and 30%_{vol}, respectively. Moreover, the authors tested NO_x emission for the different configurations of combustor and showed that NO_x emission for the air/fuel equivalence ratio above 1.3 is below 10 ppm for the dry composition. NO_x close to zero is observed at the air/fuel equivalence ratio above 1.4.

To continue the evaluation of NO_x emissions in gas turbines fired with ammonia-hydrogen blends, Aalrebei et al. developed an algorithm

for predicting the combustion and emission performances of industrial and experimental gas turbines [60]. The algorithm was realized through a code in LabVIEW [61] and GASEQ [62]. Based on the developed code, the authors tested various operating parameters and determined that extremely high NO_x emissions are observed when an ammonia-hydrogen blend is reacted with air at an equivalence ratio from 0.5 to 1.25. For equivalence ratios above 1.25, NO_x emissions are at an appropriate level.

Bovzo et al. [63] presented study on fuel rich ammonia-hydrogen injection for humidified gas turbines. It is well known that steam injection in gas turbines is one of the most widely used methods to reduce NO_x emission. Moreover, steam injection is a way to improve the energy efficiency of gas turbines. Bovzo et al. assessed the effect of steam injection on combustion and performances of gas turbine. The authors measured that NO_x monotonically decrease with an increase in the mass flow rate of injected steam. NO_x emission is achieved level of below 10 ppm. Moreover, an increase in pressure leads to decrease in NO_x emission. Thus, high pressure ammonia fired gas turbines is perspective to achieve ultra low NO_x emission. Bovzo et al. [64] also reported about investigation of humidified gas turbine fired with ammonia/hydrogen blends. The authors showed that such configuration of gas turbine can achieve the efficiency up to 60%. In addition, they evaluated pollutant emissions such as NO_x and N₂O. Steam injection in a combustion chamber leads to a significant decrease in pollutant concentration in the flue gases after gas turbine.

The papers described and analyzed above show that adding hydrogen to ammonia is a prospective way to increase the efficiency of ammonia-fired gas turbines. Moreover, ammonia-hydrogen blends are carbon-free fuel. However, the use of hydrogen as a fuel is still facing many challenges because the hydrogen supply infrastructure is not developed worldwide [65–68]. Among the different challenges, the following ones should be highlighted: transportation and storage - H₂ is pressurized up to 600 bar in a gaseous state, which requires significant modernization of the fuel supply infrastructure (hydrogen compression and transportation require a significant amount of energy) [69–71]; high cost and moderate low efficiency of “green” hydrogen production – the efficiency of various types of commercial electrolysis reactors (Acid electrolyzed water – AEW) is less than 70% [72,73]. Taking into account these challenges, it is possible to single out some ways to improve the efficiency of ammonia combustion: the use of ammonia-methane and ammonia-synthesis gas blends, or on-board hydrogen production systems.

The above-mentioned results showed that hydrogen positively affects combustion and emission performances. However, the use of hydrogen as the second co-fuel will lead to new challenges. One reason is that the gas turbine will need two fuel systems and two control systems. To avoid the addition of new fuel supply infrastructure, on-board hydrogen production via ammonia decomposition can be considered a promising technology. Section 4 of this paper focuses on this perspective approach for gas turbine applications.

3.3. Ammonia-methane blends

The research studies on combustion of ammonia-methane blends mainly focus on reactivity and flame speed, narrow flammability limits, and NO_x emissions [44,74–76]. In particular, ongoing research aims to further improve the efficiency and reduce the emissions of ammonia-methane gas turbines. Some challenges include the need for better combustion processes and the development of more durable materials to withstand the corrosive nature of ammonia. Despite these challenges, the potential benefits of this technology make it an exciting area of research for the future of sustainable energy.

There is a set of papers investigating micro-gas turbines fueled with NH₃-CH₄ blends. Between 2017 and 2019, researchers at Tohoku University developed a micro gas turbine that can operate on a mixture of ammonia and methane fuels. The micro gas turbine was found to

have high thermal efficiency and low emissions, making it a promising technology for small-scale power generation applications. The investigation also showed that the micro gas turbine fired with ammonia blends could operate stably over a wide range of operating conditions, including different fuel mixtures and load conditions [41,77,78].

Kurata et al. [77] investigated the efficiency and emission performance of micro gas turbines fueled with ammonia–methane blends. They tested the micro gas turbine under various operating conditions up to 35 kW_e to understand the effect of fuel composition on NO, NO₂, N₂O emissions, and unburnt NH₃. The results showed that NO emissions significantly increased when the methane volume ratio decreased. For a blend composition of NH₃ = 65%_{vol} and CH₄ = 35%_{vol}, a further increase in ammonia volume fraction led to an increase in unburnt NH₃ and a decrease in NO emissions. Additionally, the authors found that NO₂ emissions were not significantly affected by blend composition, with NO₂ concentrations around 40 ppm for ammonia/methane ratios (mol) from 0 to 1. Furthermore, Kurata et al. established that methane concentration had a significant effect on N₂O emissions. For methane mole fractions below 0.5, N₂O emissions were less than a few ppm. However, a further increase in ammonia mole fraction resulted in a dramatic increase in N₂O emissions. N₂O emissions were 12 ppm for 65% NH₃, while for 85% NH₃, N₂O emissions were 82 ppm. In conclusion, the authors stated that adding methane up to 35%_{vol} resulted in a significant decrease in nitrogen oxide emissions in gas turbines.

Researchers at Cardiff University tested NH₃–CH₄ combustion in tangential swirl combustor of gas turbine [79]. The authors showed that methane addition leads to notable decrease in NO_x emission, especially for the air/fuel equivalence ratio above 1. They confirmed that for the equivalence ratio about 1.5, NO_x emission is about 10 ppm what is an appropriate level for industrial application. Based on this experimental work, Xiao et al. conducted the numerical investigation of ammonia–methane combustion under gas turbine operating conditions [80]. The authors reported that an increase in pressure leads to a significant decrease in NO_x emission even at moderate low pressure. These results showed that addition of methane to ammonia in gas turbines which is operating under pressure up to 30–50 bar will lead to a notable decrease in NO_x emission.

Avila et al. [81] experimentally assessed the performance of a commercial micro gas turbine (Ansaldo AE-T100) fueled by NH₃–CH₄ blends. This gas turbine has the following parameters: power output up to 100 kW_e, pressure ratio of 5, efficiency of 30%, and turbine inlet temperature of 645 °C. Initially, this turbine was developed for natural gas combustion, but the authors added ammonia to the fuel. The experiments showed that stable operation and combustion of the gas turbine were observed until the ammonia mole fraction reached 63%. Moreover, the authors measured the concentrations of different pollutants in the flue gas. The concentration of CO and unburned CH₄ is not affected by ammonia mole fraction until 22%. However, a further increase in ammonia mole fraction leads to a significant increase in CO and unburned CH₄ concentration. Thus, it can be concluded that an increase in ammonia mole fraction results in a decrease of combustion efficiency, which illustrates the lower thermal efficiency. For example, the thermal efficiency for the blend with 63% ammonia is 23%, while the thermal efficiency of the gas turbine fueled with 100% CH₄ is 27%. Additionally, the authors showed that an increase in methane mole fraction in the fuel leads to a decrease in NO_x emissions.

Ilbas and co-authors [82] conducted an experimental investigation of the combustion stability and NO_x emissions of NH₃/CH₄ blends. The authors reported that an increase in ammonia mole fraction in the fuel leads to an increase in combustion instability. Moreover, they provided photos of flames where it is clear that the addition of NH₃ leads to an increase in flame length. Fig. 6 shows the effect of adding ammonia to methane on flame length. This figure makes it clear that the flame for combustion of 100% CH₄ is shorter and thicker than the flames for combustion of ammonia–methane blends. Therefore, these contours illustrate one of the most important issues for the application

of ammonia–methane blends as a fuel — low combustion efficiency. Another way to increase combustion efficiency is to add synthesis gas (syngas) to ammonia for combustion in gas turbines. Syngas consists mainly of hydrogen and carbon monoxide, and therefore, combustion performance of ammonia–syngas blends should be expected between ammonia–hydrogen and ammonia–methane blends. Further research is needed to optimize the use of ammonia–methane blends and explore their potential as a sustainable fuel source.

The overview of the use of ammonia–methane fueled gas turbines shows that such blends are appropriate for existing gas turbines. The combustion and emission performances of such gas turbines are close to methane-fueled gas turbines. However, this might meet with new challenges such as two fuel systems and two control systems being needed. Therefore, one of the best strategies for ammonia-fueled gas turbines is to find a solution to achieve the best combustion and emission performances without significant modification of the existing fuel supply infrastructure. Section 4 of the present paper deals with on-board hydrogen production from ammonia. This technology does not require the addition of new fuel supply and control infrastructure.

3.4. Ammonia–syngas blends

Gas turbines fired with ammonia–synthesis gas blends are a promising option for several reasons [83–86]. Firstly, synthesis gas can be produced from organic biomass without the use of fossil fuels. Secondly, the addition of syngas to ammonia can improve combustion stability and reduce emissions of NO_x, CO, and unburned hydrocarbons. Finally, gas turbines fired with ammonia–syngas blends can be retrofitted to use coal as a fuel. However, there is limited research available on gas turbines fired with ammonia–synthesis gas blends. Nonetheless, some studies have investigated the combustion performance and emissions of these blends.

Cai et al. [87] investigated the effects of ammonia–syngas blends on combustion stability, emissions, and thermal efficiency in a gas turbine combustor. The authors reported that the addition of syngas to ammonia improved the combustion stability and reduced NO_x emissions. They also found that increasing the syngas fraction in the blend led to a decrease in CO and unburned hydrocarbon emissions. However, the thermal efficiency of the combustor was slightly reduced with the addition of syngas.

Another study by Han et al. [88] investigated the combustion characteristics and emissions of ammonia–syngas blends in a swirl-stabilized combustor. The authors reported that increasing the syngas fraction in the blend led to a decrease in flame temperature and NO_x emissions. However, they also found that the addition of syngas increased CO and unburned hydrocarbon emissions.

Mei et al. [89] investigated an effect of synthesis gas addition to ammonia for the pressure range up to 10 bar which is corresponding to conditions in gas turbines. The authors was mainly focused on the study of effect of blend composition on NO_x emission and flame speed. They confirmed the results observed for the combustion of ammonia–hydrogen and ammonia–methane blends. An increase in mole fraction of syngas leads to decrease in NO_x emission. Moreover, they established that syngas as a blend component gives higher laminar flame speed than methane.

There is a set of papers where considered gas turbines fired with ammonia based blends with other various fuels such as methanol [11, 90], ethanol [91,92], dimethyl ether [93], etc. Kagan Ayaz et al. provided review on the use of various ammonia based blends in gas turbines [24]. For example, they considered ammonia–methanol blend as a carbon free fuel because methanol can be produced from CO₂ using renewable energy.

One example is a research project conducted by the National Renewable Energy Laboratory (NREL) in the United States [94]. The project involved testing a gas turbine fueled with a blend of ammonia and dimethyl ether (DME), which is a synthetic fuel made from natural

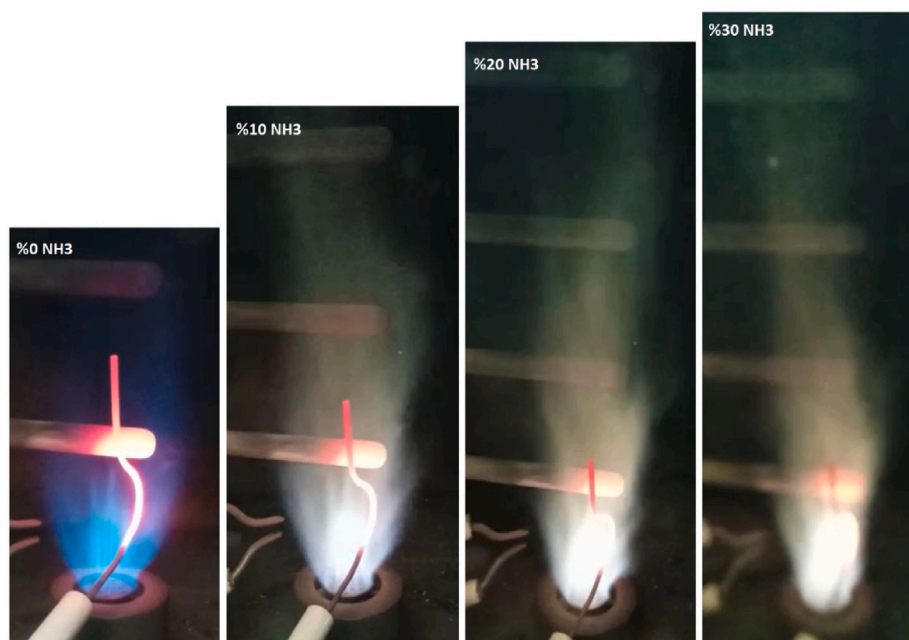


Fig. 6. Effect of addition of ammonia to methane on the flame length [82].

gas or biomass. The results showed that the ammonia-DME blend had lower emissions of NO_x, CO, and unburned hydrocarbons compared to conventional natural gas fuels.

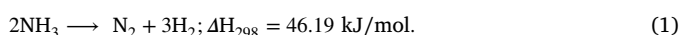
Despite recent advances in the development of the use of renewable energy sources, there are some countries where coal is one of the main fuels for power generation [95–97]. However, coal-fired power plants have extremely high emissions [98–100]. To avoid negative effect of coal-fired power plant, an integrated gasification combined cycle (IGCC) is used to reduce CO₂ emissions [101–103].

Among existing gas turbines fired with ammonia-syngas blends, Chugoku Electric Power Company (Japan) should be highlighted [104]. This company tested a gas turbine fired with ammonia-syngas at Mizushima power plant with power output about 150 MW. Gas turbine fired with synthesis gas after gasification was co-fueled with ammonia. Chugoku Electric Power Company reported that adding ammonia have no significant effect on power efficiency and power output. However, the reduction of pollutant emission was obtained for NO_x and CO₂ emissions of the ammonia did not cause the plant's power efficiency to reduce.

The integrated gasification combined cycle (IGCC) can be considered as an on-board synthesis gas production system. In other words, for such plants, there is no need to have a developed fuel supply infrastructure. Syngas is produced on-board before its utilization as a fuel. As shown above, the most promising combustion and emission performances of ammonia-fired gas turbines occur when ammonia-hydrogen blends are used. However, to use hydrogen as a fuel, the challenges of hydrogen supply infrastructure must be overcome. One way to solve the problem of hydrogen supply is an onboard hydrogen production system integrated into gas turbines [105].

4. On-board ammonia thermochemical transformation

One way to solve the challenges related to hydrogen fuel supply infrastructure is to produce hydrogen before its utilization in gas turbines. Ammonia is a highly efficient hydrogen carrier and can generate the highest hydrogen mass per unit mass of carrier among various environmentally friendly hydrogen carriers [106–108]. Hydrogen extraction from ammonia can be achieved through a reaction of ammonia decomposition [109,110].



The products of such a transformation are a mixture of nitrogen and hydrogen. Therefore, after ammonia decomposition, combustion of the hydrogen-rich mixture will take place in fuel-consuming equipment. Compared to ammonia, hydrogen is a fuel with a high flame speed and short quenching distance, leading to enhanced and varied combustion within gas turbines. The ammonia thermochemical transformation (ATT) process is an endothermic process, and external heat sources must be supplied to convert ammonia into hydrogen-rich gas. Heat sources for ammonia thermochemical transformation include:

- electric heating, which provides a clean and efficient source of heat [111,112];
- ammonia combustion to provide heat for the thermochemical transformation process [113,114];
- waste heat from gas turbines generating exhaust heat with temperatures up to 500–600 °C, which can be used to supply external heat for the decomposition process [115,115];
- concentrated solar energy, which can provide the high temperatures required for ammonia thermochemical transformation [116, 117]

The last two sources mentioned above are the most promising for application in gas turbines with ammonia thermochemical transformation because they use renewable and waste energy for NH₃ decomposition. Recently, several papers have been published on the use of exhaust heat for thermochemical transformation of various fuels such as natural gas [118,119], ethanol [120,121], and methanol [122,123]. This use of exhaust heat for thermochemical fuel transformation is called thermochemical recuperation [124–127].

Thermochemical waste heat recuperation is a process that involves the recovery of waste heat from industrial processes and converting it into useful energy through thermochemical reactions. This process typically involves the use of high-temperature heat sources, such as exhaust gases from furnaces or gas turbines, to drive chemical reactions that produce useful products, such as hydrogen or synthetic fuels. Thermochemical waste heat recuperation can improve the energy efficiency of industrial processes and reduce greenhouse gas emissions by capturing waste heat that would otherwise be lost. It is an important area of research for developing sustainable and efficient energy systems.

The product of thermochemical ammonia transformation is hydrogen-rich gas which can be used as a fuel in gas turbine. Therefore,

the concept of on-board hydrogen production is observed in this case. Hydrogen-rich gas produced via ATT can be a viable fuel for gas turbines, with comparable performance to pure hydrogen combustion. Hydrogen-rich fuel combustion produces nitrogen and water as byproducts, which are environmentally friendly compared to the greenhouse gases produced by fossil fuels.

There are some recently published papers on ammonia-fired gas turbines with thermochemical ammonia transformation, or ammonia-fired chemically recuperated gas turbines. One of the first papers on this topic was presented by Pashchenko and co-authors [115]. In this paper, the authors analyzed the effect of operating parameters on the efficiency of a gas turbine when exhaust heat is used for thermochemical ammonia transformation. The results showed that the maximum increase in efficiency is about 12% when comparing a gas turbine with and without thermochemical ammonia transformation.

Su et al. [128] proposed a novel ammonia-driven chemically recuperated gas turbine cycle based on a dual-fuel mode. They suggested using exhaust heat for ammonia evaporation and thermochemical transformation. The authors determined the optimal ratio of ammonia transformation, and the results showed that the efficiency of the proposed cycle is about 35.48–36.09%. This efficiency is notably higher than the efficiency of a gas turbine without thermochemical ammonia transformation.

Moreover, the use of renewable energy sources is a promising way for on-board hydrogen production from ammonia. However, there are no papers on integrated ammonia-fired gas turbines with ammonia transformation by renewable heat sources, such as solar energy. Therefore, integrating an ammonia-fired gas turbine with a solar energy system will have a double advantage: the use of zero-carbon fuel and a renewable energy source.

Research papers on ammonia-fired gas turbines with thermochemical ammonia transformation are quite limited. In this section, the concept of a chemically recuperated ammonia-fired gas turbine (CRGT) and an integrated solar ammonia-fired gas turbine (ISGT) from the point of view of the thermal efficiency of gas turbines is analyzed.

4.1. Ammonia fired chemically recuperated gas turbines

The concept of an ammonia-fueled chemically recuperated gas turbine is based on using exhaust heat for the thermochemical transformation of ammonia. In this turbine, the conversion of chemical energy from ammonia into heat is divided into two stages. The first stage involves decomposing ammonia to obtain a hydrogen-rich gas. It should be noted that the lower heating value of 1 kg of hydrogen-rich gas is higher than the lower heating value of 1 kg of initial ammonia. The second stage involves the combustion of the hydrogen-rich gas. The reaction for thermochemical transformation of ammonia is highly endothermic, which means that external heat needs to be supplied to the reaction mixture. According to Eq. (1), 46.19 kJ/mol is required to transform 1 mole of ammonia into a hydrogen–ammonia mixture.

The schematic diagram of ammonia fueled gas turbine with thermochemical ammonia transformation is shown in Fig. 7. Exhaust gas after gas turbine is schematic passing through an ammonia reformer and evaporator. In the ammonia reformer, the reaction of ammonia decomposition is occurring according to Eq. (1). Hydrogen-rich gas is used as a fuel for gas turbine. Exhaust gas after the ammonia reformer is used to evaporate ammonia because ammonia at ambient temperature of 15–25 °C and pressure above 10 bar is a liquid.

To analyze the thermal efficiency of the gas turbine presented in Fig. 7, a thermodynamic analysis was performed. The analysis of this gas turbine with thermochemical ammonia transformation can be virtually divided into two steps: the first step is the thermodynamic analysis of the ammonia decomposition process, and the second step is the thermodynamic analysis of the cycle. According to Le-Chatelier's principle, the ammonia decomposition process is strongly dependent on operating parameters, such as temperature and pressure, because the

process is endothermic and proceeds with an increase in the number of moles. For both steps of thermodynamic analysis, Aspen HYSYS software is used. Aspen HYSYS is a powerful process simulation software that can be used for thermodynamic analysis of chemically recuperated gas turbines. It has a wide range of capabilities, including the ability to model complex chemical reactions and calculate thermodynamic properties of fluids and mixtures. Aspen HYSYS also allows for the simulation of various operating conditions, such as temperature and pressure, which are important parameters in the thermodynamic analysis of chemically recuperated gas turbines. Additionally, Aspen HYSYS can be used to optimize the design and operation of gas turbine systems, allowing for the identification of potential areas for improvement in terms of thermal efficiency and overall performance. Therefore, Aspen HYSYS can be considered as a valuable tool for the thermodynamic analysis of chemically recuperated gas turbines [129,130].

The computational scheme of ammonia fired gas turbine from Fig. 7 realized via Aspen HYSYS is presented in Fig. 8. The computational scheme consist of five elements: evaporator to evaporate liquid ammonia before the decomposition process, ammonia reformer where ammonia decomposition is taking place, compressor, combustion chamber, and turbine. The ammonia reformer set as a Gibbs reactor. The isentropic efficiency of compressor and turbine is 0.85. The reference value of the net power output of gas turbine unit is 40 MW. This power output is corresponding to the latest ammonia fired gas turbine produced by Mitsubishi Corp. [25].

In general, the thermal efficiency of a simple gas turbine depends on the following operational parameters: pressure ratio and turbine inlet temperature. However, the thermal efficiency of a chemically recuperated gas turbine also depends on the degree of ammonia decomposition, which is affected by pressure and temperature. The degree of ammonia decomposition, as well as the composition of the hydrogen-rich gas after a reformer, depends on temperature and pressure. Fig. 9 shows the effect of pressure and temperature on the degree of ammonia decomposition (a) and hydrogen-rich gas composition (b). To verify the thermodynamic model from Fig. 8, experimental results on ammonia decomposition obtained at the Fixed Nitrogen Research Laboratory (F.N.R.L) were used [131]. Fig. 9a depicts good agreement between numerical and experimental results. As shown in Fig. 9a, an increase in pressure leads to a decrease in ammonia decomposition, and an increase in temperature leads to an increase in ammonia decomposition. These facts are fully consistent with Le Chatelier's principle. Moreover, Fig. 9b illustrates the effect of temperature and pressure on the equilibrium hydrogen-rich gas composition. An increase in process temperature increases the hydrogen molar fraction, and an increase in pressure leads to a decrease in hydrogen mole fraction. Fig. 9b shows that at a temperature of 400 °C, the hydrogen-rich gas mainly consists of hydrogen (up to 75%). Therefore, the products of ammonia decomposition can be considered as a hydrogen-rich fuel for gas turbines. It should be noted that Fig. 9 depicts the equilibrium gas compositions, which depend only on pressure and temperature. In existing reformers, the composition of hydrogen-rich gas also depends on the type of catalyst, mass of catalyst, mass flow rate, residence time, design parameters, etc. However, there is sufficient information in the literature that modern reformers can operate under conditions close to equilibrium [132–134].

Fig. 10 illustrates the impact of turbine inlet temperature (700–1000 °C) and pressure ratio (6–18) on the thermal efficiency of an ammonia-fired gas turbine with (a) and without (b) thermochemical recuperation. The results demonstrate that utilizing exhaust heat for the thermochemical transformation of ammonia leads to a significant increase in thermal efficiency. The average efficiency increase is up to 5%–7% for various operational parameters. Additionally, Fig. 10a indicates that the thermochemical exhaust heat recuperation system decreases the optimal pressure ratio. Therefore, the maximum efficiency of an ammonia-fired gas turbine can be achieved at a lower pressure ratio. For instance, the optimal pressure ratio for ammonia-fired gas

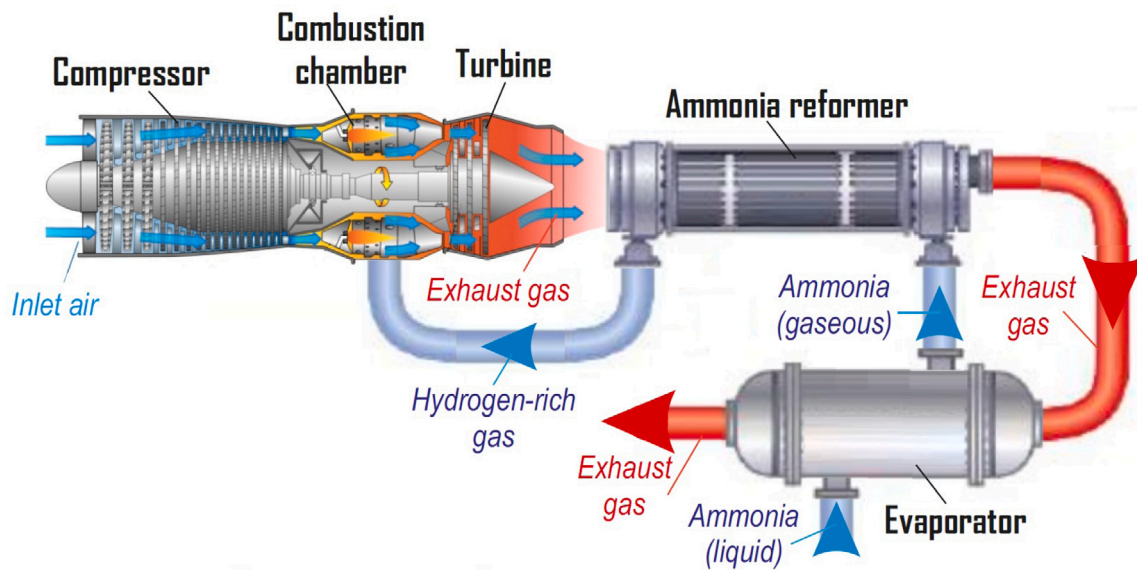


Fig. 7. Schematic diagram of ammonia fueled gas turbine with thermochemical transformation of ammonia using exhaust heat.

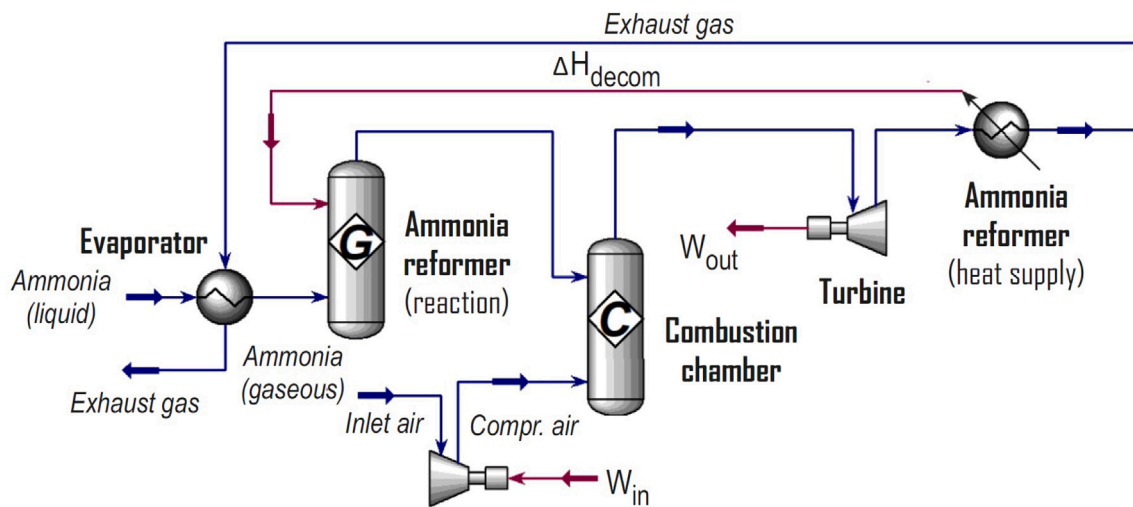


Fig. 8. The computational scheme of ammonia fired gas turbine from Fig. 7 realized via Aspen HYSYS.

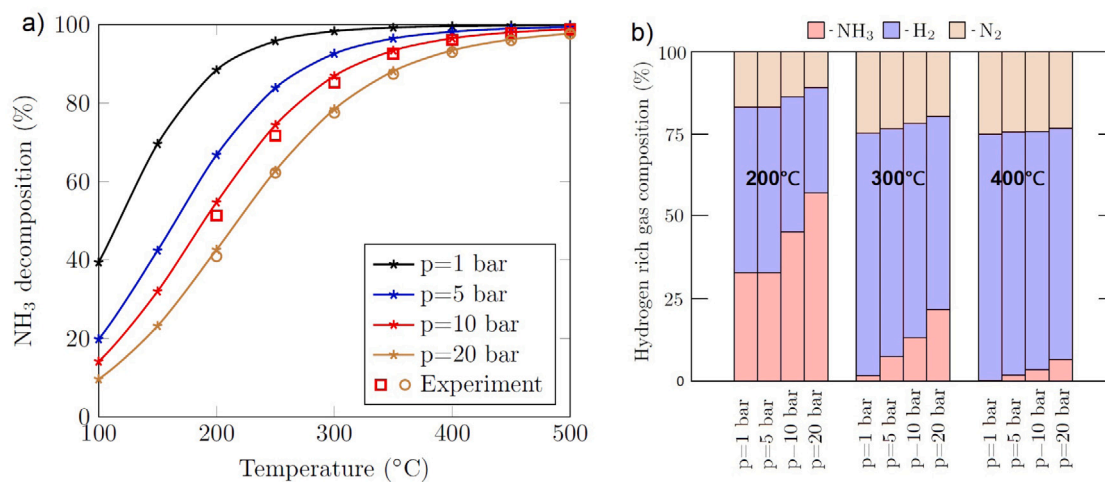


Fig. 9. Equilibrium ammonia decomposition degree as a function of temperature and pressure: solid lines – equilibrium decomposition from the present thermodynamic analysis; hollow points – experimental data from [131].

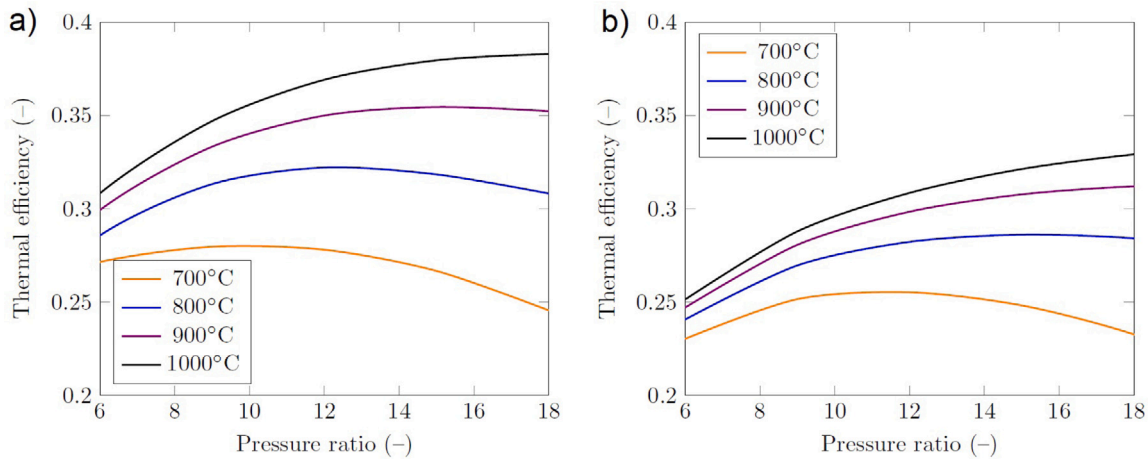


Fig. 10. The thermal efficiency of ammonia fired gas turbine with thermochemical recuperation (a) and without thermochemical recuperation (b) as a function of turbine inlet temperature and pressure ratio.

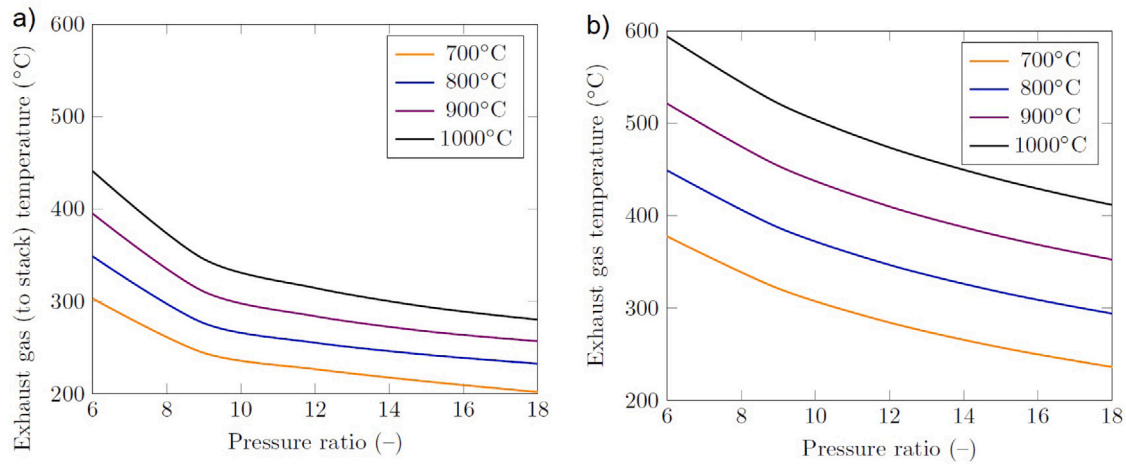


Fig. 11. Temperature of exhaust gas (to chimney) for various pressure ratio and turbine inlet temperature after gas turbine with thermochemical recuperation (a) and without thermochemical recuperation (b).

turbines with and without thermochemical recuperation is 10.3 vs. 11.8 for TIT = 700 °C, 12.2 vs. 14.9 for TIT = 800 °C, and 14.4 vs. 21.2 for TIT = 900 °C.

Fig. 10 clearly indicates that the use of thermochemical recuperation is an effective engineering solution. In addition, in this case, hydrogen-rich gas with a molar fraction of up to 75% is used as fuel in the combustion chamber. Based on the literature review in Section 3, this fuel significantly increases combustion stability and results in lower emissions compared to the combustion of pure ammonia.

The use of thermochemical exhaust recuperation in ammonia-fired gas turbines results in a significant increase in thermal efficiency. This is achieved by converting exhaust heat into chemical energy in the form of hydrogen-rich gas, which has a higher lower heating value than pure ammonia. As a result of heat recuperation, the temperature of the exhaust gas is reduced. Fig. 11 illustrates the effect of thermochemical recuperation on exhaust gas temperature for various pressure ratios and turbine inlet temperatures. The figure shows that the thermochemical recuperation system effectively reduces exhaust gas temperature, making it suitable for adding either a low-temperature steam turbine cycle or an Organic Rankine Cycle (ORC). Therefore, the study of combined power cycle plants based on ammonia-fired gas turbines with thermochemical recuperation is a promising approach to increasing the efficiency of power generation systems.

4.2. Integrated solar ammonia fired gas turbine

Another option for obtaining hydrogen-rich fuel from ammonia is through the utilization of renewable solar energy. The temperature potential of solar energy from solar fields is suitable for the ammonia decomposition process [135–137]. As shown in Fig. 9, to achieve ammonia transformation close to unity (approximately 100% converted to hydrogen-rich gas), a moderate temperature level of around 400–500 °C is required. In this case, solar energy is used as a heat source for the ammonia decomposition reaction (1). The use of solar energy as a heat source offers several advantages:

- Renewable and sustainable: Solar energy is a renewable and sustainable source of energy, which means that it can be used indefinitely without depleting natural resources.
- Clean and environmentally friendly: Solar energy does not produce any greenhouse gas emissions or air pollutants, making it a clean and environmentally friendly source of energy.
- Cost-effective: The cost of solar energy has been decreasing over the years, making it a cost-effective alternative to traditional fossil fuels.
- No need high concentrated solar power systems: ATT is occurring at the moderate temperature levels and no need high concentrated solar power systems to generate the high-temperature energy sources.

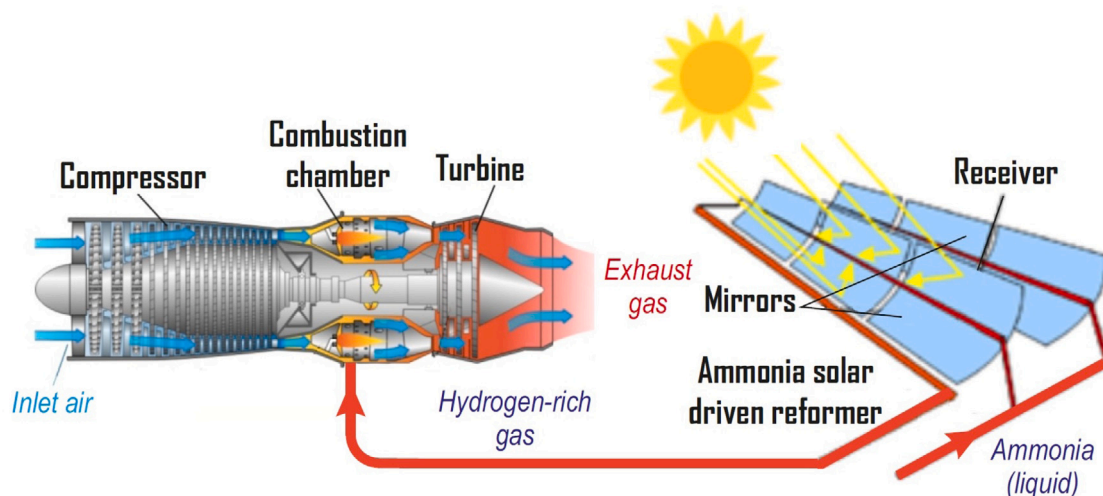


Fig. 12. Schematic diagram of integrated solar ammonia fired gas turbine.

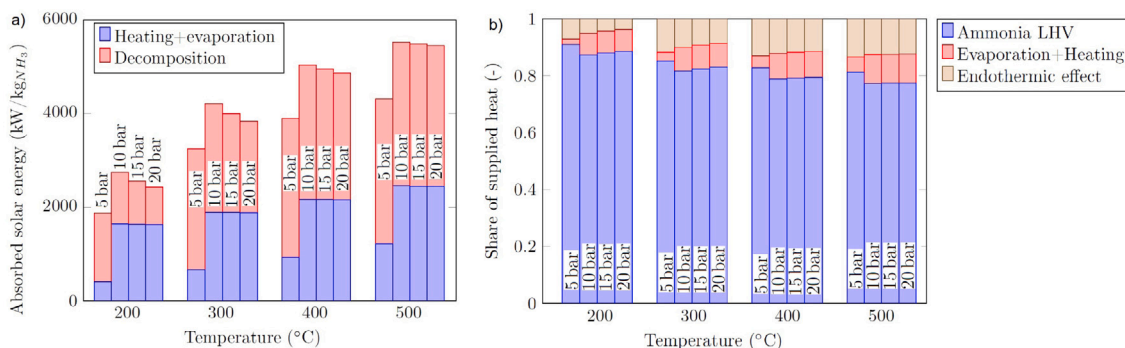


Fig. 13. Heat absorbed by solar receiver per 1 kg/s of ammonia (a) and heat balance of combustion chamber of gas turbine (b).

The use of solar energy for ammonia decomposition will lead to an increase in the exhaust gas temperature. Therefore, this approach will increase the efficiency of combined power cycles based on ammonia-fired gas turbines with ammonia thermochemical transformation. According to the average electrical load schedule, the maximum electricity consumption takes place during sunny days. Therefore, using solar energy for ammonia transformation can cover additional energy consumption during the day. Moreover, the ammonia thermochemical transformation process can be considered a thermochemical solar energy storage system where solar energy is stored in the form of hydrogen-rich gas.

The schematic diagram of an integrated solar ammonia-fired gas turbine is presented in Fig. 12. In this gas turbine, ammonia is decomposed before combustion to obtain hydrogen-rich gas. Solar energy is utilized as the heat source for this endothermic process. As a result of this transformation, the combustion chamber is fueled with gas fuel where the molar fraction of hydrogen is about 75%. Solar energy is supplied to evaporate, preheat, and decompose ammonia.

To understand the impact of operational parameters on the heat balance of the ammonia decomposition system, a thermodynamic analysis was conducted. Aspen HYSYS was used for the analysis, and the computational scheme was similar to that shown in Fig. 8, except that heat for ammonia evaporation, heating, and decomposition was supplied from an external heat source. Fig. 13 illustrates the amount of heat absorbed by the solar receiver per 1 kg/s of ammonia (a) and the heat balance of the combustion chamber of the gas turbine (b).

Fig. 13a shows the amount of heat absorbed by 1 kg/s of ammonia in a solar receiver for different pressures and temperatures. As can be seen, the minimum amount of heat absorbed in the solar receiver

is at a pressure of 5 bar. This is because at a pressure of 5 bar and a temperature of 25 °C, ammonia is in a gaseous state and no heat flow for evaporation is observed. The maximum heat absorbed in the receiver is at a pressure of 10 bar for all temperatures because in this case, the maximum heat supplied for evaporation is observed as well as the maximum degree of ammonia decomposition. According to Fig. 9a, an increase in pressure leads to a decrease in ammonia decomposition degree.

Moreover, Fig. 13b illustrates the heat balance of the combustion chamber. As can be seen, the additional heat flow from a solar energy supply system increases with an increase in temperature in the receiver. The maximum heat flow from solar energy is observed at a temperature of 500 °C, which is achievable for modern mirror solar receivers [138–140]. For a temperature of 500 °C and a pressure of 20 bar, approximately 23% of the required heat for the gas turbine is supplied from a solar energy system. This means that 23% of the heat from initial ammonia can be replaced by solar energy. Additionally, in this case, the hydrogen-rich gas will be used as fuel in the gas turbine, and the advantages of this fuel will be applied for the integrated solar ammonia-fired gas turbine.

The thermodynamic analysis indicated that the exhaust gas temperature after the integrated solar ammonia-fired gas turbine is the same as for the simple gas turbine presented in Fig. 11b. Therefore, it can be concluded that there is enough temperature potential for integration of either a low-temperature steam turbine cycle or an Organic Rankine Cycle (ORC). The study of combined solar power generation systems is a promising way to develop carbon-free, highly efficient power generation systems. Additionally, it should be noted that the cost of heat produced from ammonia is about 30% higher than the cost of

heat produced by a solar mirror receiver []. Therefore, studying such systems from a techno-economic point of view is a crucial engineering task because in this case, the use of solar energy for ammonia transformation will increase the economic efficiency of using ammonia in a gas turbine as fuel.

5. Conclusion

Ammonia is considered a promising fuel for the transition to carbon-free energy because it can be used in existing gas turbines without significant modification of the schematic diagram. However, there are challenges in the use of ammonia as a fuel. The main ones are increased NO_x emissions, unstable combustion, and decreased thermal efficiency. In this paper, various aspects of the use of ammonia as a gas turbine fuel are considered. The existing gas turbines fired with pure ammonia are reviewed. To solve the problems in the use of pure ammonia as a gas turbine fuel, adding other combustible components such as hydrogen, methane, and synthesis gas is considered. The combustion properties (adiabatic temperature, flame speed, lower heating value) are calculated for different compositions of ammonia–hydrogen blends. It is shown that the addition of hydrogen to ammonia has an insignificant effect on combustion properties if the hydrogen mole fraction is less than 50% because the density of hydrogen is about six times less than the density of ammonia. However, the addition of hydrogen up to 20%_{mol} leads to a decrease in NO_x emissions and an increase in combustion stability. Moreover, an increase in pressure leads to a decrease in NO_x emissions. For example, a blend of $\text{NH}_3:\text{H}_2 = 5:1$ (mol/mol) gives NO_x emissions less than 12 ppm at a pressure of 5 bar. This level of NO_x emission corresponds to the NO_x emissions of existing gas turbines fired with natural gas. In addition, the use of ammonia–methane blends as a gas turbine fuel is reviewed. The addition of methane to ammonia leads to a slight change in flame speed, adiabatic temperature, and lower heating value. However, stable combustion of ammonia–methane blends is observed at CH_4 mole fractions above 23%. Therefore, the addition of methane increases combustion stability. There are a few existing gas turbines where ammonia–methane blends are used as fuel without significant modification of the schematic diagram of the gas turbine. Another ammonia-containing blend considered in this paper is ammonia–synthesis gas blends. These blends can be considered a perspective way for the use of ammonia in existing integrated gasification combined cycles fueled with coal.

The second part of this paper presents the findings of a study on a gas turbine equipped with on-board ammonia thermochemical transformation prior to combustion. Two methods were examined: an ammonia-fired gas turbine with thermochemical exhaust heat recovery, and an integrated solar ammonia-fired gas turbine. The thermal efficiency of both gas turbines was assessed. The utilization of exhaust heat for ammonia decomposition resulted in a rise in thermal efficiency of up to 5%–7% for various operating conditions. The maximum efficiency of an ammonia-fired gas turbine can be achieved at a lower pressure ratio. The optimal pressure ratio for ammonia-fired gas turbines with and without thermochemical recuperation is 10.3 vs. 11.8 for TIT = 700 °C, 12.2 vs. 14.9 for TIT = 800 °C, and 14.4 vs. 21.2 for TIT = 900 °C. Additionally, ammonia transformation generates hydrogen-rich gas (up to 75% hydrogen) with superior combustion performance compared to pure ammonia. Thus, thermochemical exhaust heat recovery through ammonia decomposition enhances both thermal efficiency and combustion stability. The second method examined in this paper is an integrated solar ammonia-fired gas turbine where solar energy is utilized to compensate for the endothermic effect of ammonia decomposition. The thermodynamic analysis revealed that the use of solar energy can lower ammonia consumption by up to 25% by replacing heat generated through ammonia combustion with heat from the solar receiver. Furthermore, the future prospects of utilizing ammonia thermochemical transformation before combustion are discussed.

CRediT authorship contribution statement

Dmitry Pashchenko: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] Sovacool Benjamin K, Griffiths Steve. The cultural barriers to a low-carbon future: A review of six mobility and energy transitions across 28 countries. *Renew Sustain Energy Rev* 2020;119:109569.
- [2] Perlaviciute Goda, Steg Linda, Sovacool Benjamin K. A perspective on the human dimensions of a transition to net-zero energy systems. *Energy Clim Change* 2021;2:100042.
- [3] Slameršak Aljoša, Kallis Giorgos, O'Neill Daniel W. Energy requirements and carbon emissions for a low-carbon energy transition. *Nat Commun* 2022;13(1):6932.
- [4] Yu Bolin, Fang Debin, Xiao Kun, Pan Yuling. Drivers of renewable energy penetration and its role in power sector's deep decarbonization towards carbon peak. *Renew Sustain Energy Rev* 2023;178:113247.
- [5] Zhang Dahai, Wang Jiaqi, Lin Yonggang, Si Yulin, Huang Can, Yang Jing, Huang Bin, Li Wei. Present situation and future prospect of renewable energy in China. *Renew Sustain Energy Rev* 2017;76:865–71.
- [6] Shuai Jing, Zhao Yujia, Wang Yilan, Cheng Jinhua. Renewable energy product competitiveness: Evidence from the United States, China and India. *Energy* 2022;249:123614.
- [7] Bildirici Melike, Kayıkçı Fazıl. Renewable energy and current account balance nexus. *Environ Sci Pollut Res* 2022;29(32):48759–68.
- [8] Zamfirescu C, Dincer I. Using ammonia as a sustainable fuel. *J Power Sources* 2008;185(1):459–65.
- [9] Herbinet Olivier, Bartocci Pietro, Dana Alon Grinberg. On the use of ammonia as a fuel—A perspective. *Fuel Commun* 2022;11:100064.
- [10] Chai Wai Siong, Bao Yulei, Jin Pengfei, Tang Guang, Zhou Lei. A review on ammonia, ammonia–hydrogen and ammonia–methane fuels. *Renew Sustain Energy Rev* 2021;147:111254.
- [11] Blanco Elena C, Sánchez Antonio, Martín Mariano, Vega Pastora. Methanol and ammonia as emerging green fuels: Evaluation of a new power generation paradigm. *Renew Sustain Energy Rev* 2023;175:113195.
- [12] Chatterjee Sudipta, Parsapur Rajesh Kumar, Huang Kuo-Wei. Limitations of ammonia as a hydrogen energy carrier for the transportation sector. *ACS Energy Lett* 2021;6(12):4390–4.
- [13] Cardoso João Sousa, Silva Valter, Rocha Rodolfo C, Hall Matthew J, Costa Mário, Eusébio Daniela. Ammonia as an energy vector: Current and future prospects for low-carbon fuel applications in internal combustion engines. *J Clean Prod* 2021;296:126562.
- [14] Chiong Meng-Choung, Chong Cheng Tung, Ng Jo-Han, Mashruk Syed, Chong William Woei Fong, Samiran Nor Afzanizam, Mong Guo Ren, Valera-Medina Agustin. Advancements of combustion technologies in the ammonia-fuelled engines. *Energy Convers Manage* 2021;244:114460.
- [15] Dimitriou Pavlos, Javaid Rahat. A review of ammonia as a compression ignition engine fuel. *Int J Hydrogen Energy* 2020;45(11):7098–118.
- [16] Li Tie, Zhou Xinyi, Wang Ning, Wang Xinran, Chen Run, Li Shiyang, Yi Ping. A comparison between low-and high-pressure injection dual-fuel modes of diesel-pilot-ignition ammonia combustion engines. *J Energy Inst* 2022;102:362–73.
- [17] Li Jinguang, Zhang Ren, Pan Jiaying, Wei Haiqiao, Shu Gequn, Chen Lin. Ammonia and hydrogen blending effects on combustion stabilities in optical SI engines. *Energy Convers Manage* 2023;280:116827.
- [18] Valera-Medina Agustin, Xiao Hua, Owen-Jones Martin, David William IF, Bowen PJ. Ammonia for power. *Prog Energy Combust Sci* 2018;69:63–102.
- [19] MacFarlane Douglas R, Cherepanov Pavel V, Choi Jaechol, Suryanto Bryan HR, Hodgetts Rebecca Y, Bakker Jacinta M, Vallana Federico M Ferrero, Simonov Alexandr N. A roadmap to the ammonia economy. *Joule* 2020;4(6):186–205.

- [20] Liu Long, Wu Yue, Wang Yang, Wu Jie, Fu Shiyi. Exploration of environmentally friendly marine power technology-ammonia/diesel stratified injection. *J Clean Prod* 2022;380:135014.
- [21] Machaj K, Kupecki J, Malecha Z, Morawski AW, Skrzypkiewicz M, Stanclik M, Chorowski M. Ammonia as a potential marine fuel: A review. *Energy Strategy Rev* 2022;44:100926.
- [22] Sun Wanchen, Zeng Wenpeng, Guo Liang, Zhang Hao, Yan Yuying, Lin Shao-dian, Zhu Genan, Jiang Mengqi, Yu Changyou, Wu Fei. An optical study of the combustion and flame development of ammonia-diesel dual-fuel engine based on flame chemiluminescence. *Fuel* 2023;349:128507.
- [23] Ejder Emir, Arslanoğlu Yasin. Evaluation of ammonia fueled engine for a bulk carrier in marine decarbonization pathways. *J Clean Prod* 2022;379:134688.
- [24] Ayaz S Kagan, Altuntas Onder, Caliskan Hakan. Enhanced life cycle modelling of a micro gas turbine fuelled with various fuels for sustainable electricity production. *Renew Sustain Energy Rev* 2021;149:111323.
- [25] 2021. [Online; accessed 03-March-2021]. <https://power.mhi.com/news/20210301.html>.
- [26] Ito Shintaro, Uchida Masahiro, Suda Toshiyuki, Fujimori Toshiro. Development of ammonia gas turbine co-generation technology. *IHI Eng Rev* 2020;53:1–6.
- [27] Singh Dig Vijay, Pedersen Eilif. A review of waste heat recovery technologies for maritime applications. *Energy Convers Manage* 2016;111:315–28.
- [28] 2023. [Online; accessed 23-May-2023]. <https://www.man-es.com/oil-gas/products/high-pressure-vessels>.
- [29] 2023. [Online; accessed 23-May-2023]. <https://global.kawasaki.com/en/energy/>.
- [30] Emerson Benjamin, Wu David, Lieuwen Tim, Sheppard Scott, Noble David, Angello Leonard. Assessment of current capabilities and near-term availability of hydrogen-fired gas turbines considering a low-carbon future. In: *Turbo expo: power for land, sea, and air*, Vol. 84157. American Society of Mechanical Engineers; 2020, V006T09A007.
- [31] Noble David, Wu David, Emerson Benjamin, Sheppard Scott, Lieuwen Tim, Angello Leonard. Assessment of current capabilities and near-term availability of hydrogen-fired gas turbines considering a low-carbon future. *J Eng Gas Turb Power* 2021;143(4).
- [32] Shepherd Jack, Khan Muhammad Haider Ali, Amal Rose, Daiyan Rahman, MacGill Iain. Open-source project feasibility tools for supporting development of the green ammonia value chain. *Energy Convers Manage* 2022;274:116413.
- [33] Lee Hookyung, Lee Min-Jung. Recent advances in ammonia combustion technology in thermal power generation system for carbon emission reduction. *Energies* 2021;14(18):5604.
- [34] Croluis Stephen H. Ammonia's role in the hydrogen society. In: *CO2 free ammonia as an energy carrier: Japan's insights*. Springer; 2022, p. 49–61.
- [35] Verkamp FJ, Hardin MC, Williams JR. Ammonia combustion properties and performance in gas-turbine burners. In: *Symposium (international) on combustion*, Vol. 11. Elsevier; 1967, p. 985–92.
- [36] Pratt DT. Performance of ammonia-fired gas-turbine combustors. Technical report, California Univ Berkeley Thermal Systems Div; 1967.
- [37] Li Jun, Huang Hongyu, Kobayashi Noriyuki, He Zhaohong, Nagai Yoshihiro. Study on using hydrogen and ammonia as fuels: Combustion characteristics and NOx formation. *Int J Energy Res* 2014;38(9):1214–23.
- [38] Newhall HK, Starkman Ernest S. Theoretical performance of ammonia as a gas turbine fuel. *SAE Trans* 1967;772:84.
- [39] Karabeyoglu Arif, Evans Brian. Fuel conditioning system for ammonia fired power plants. In: *Proceedings of the 9th annual NH3 fuel association conference*. 2012.
- [40] Balestri M, Cecchini D, Cinti V. Unconventional fuels experimental campaigns in gas turbine combustor at ENEL sesta facility. In: *Turbo expo: power for land, sea, and air*, Vol. 41669. 2004, p. 121–8.
- [41] Kurata Osamu, Iki Norihiko, Inoue Takahiro, Matsunuma Takayuki, Tsujimura Taku, Furutani Hirohide, Kawano Masato, Arai Keisuke, Okafor Ekenechukwu Chijioke, Hayakawa Akihiro, et al. Development of a wide range-operable, rich-lean low-NOx combustor for NH3 fuel gas-turbine power generation. *Proc Combust Inst* 2019;37(4):4587–95.
- [42] Okafor Ekenechukwu C, Somarathne KD Kunkuma A, Hayakawa Akihiro, Kudo Taku, Kurata Osamu, Iki Norihiko, Kobayashi Hideaki. Towards the development of an efficient low-NOx ammonia combustor for a micro gas turbine. *Proc Combust Inst* 2019;37(4):4597–606.
- [43] Iki Norihiko, Kurata Osamu, Matsunuma Takayuki, Inoue Takahiro, Tsujimura Taku, Furutani Hirohide, Kobayashi Hideaki, Hayakawa Akihiro, Arakawa Yoshiyuki, Ichikawa Akinori. Micro gas turbine firing ammonia. In: *Turbo expo: power for land, sea, and air*, Vol. 49866. American Society of Mechanical Engineers; 2016, V008T23A018.
- [44] Ariemma Giovanni Battista, Sorrentino Giancarlo, Ragucci Raffaele, de Joannon Mara, Sabia Pino. Ammonia/methane combustion: stability and NOx emissions. *Combust Flame* 2022;241:112071.
- [45] Vigueras-Zúñiga Marco Osvaldo, Tejeda-del Cueto Maria Elena, Mashruk Syed, Kovaleva Marina, Ordóñez-Romero Cesar Leonardo, Valera-Medina Agustín. Methane/ammonia radical formation during high temperature reactions in swirl burners. *Energies* 2021;14(20):6624.
- [46] Khateeb Abdulrahman A, Guiberti Thibault F, Wang Guoqing, Boyette Wesley R, Younes Mourad, Jamal Aqil, Roberts William L. Stability limits and NO emissions of premixed swirl ammonia-air flames enriched with hydrogen or methane at elevated pressures. *Int J Hydrogen Energy* 2021;46(21):11969–81.
- [47] Yin Geyuan, Xiao Bo, Zhan Haochen, Hu Erjiang, Huang Zuohua. Chemical kinetic study of ammonia with propane on combustion control and NO formation. *Combust Flame* 2023;249:112617.
- [48] Dincer Ibrahim, Erdemir Dogan, Aydin Muhammed Iberia, Karasu Huseyin, Vezina Greg. Utilization of ammonia and ammonia blends in power generators. In: *Ammonia energy technologies*. Springer; 2023, p. 235–355.
- [49] Dinesh MH, Pandey Jayashish Kumar, Kumar GN. Study of performance, combustion, and NOx emission behavior of an SI engine fuelled with ammonia/hydrogen blends at various compression ratio. *Int J Hydrogen Energy* 2022;47(60):25391–403.
- [50] Chen Jundie, Jiang Xue, Qin Xiaokang, Huang Zuohua. Effect of hydrogen blending on the high temperature auto-ignition of ammonia at elevated pressure. *Fuel* 2021;287:119563.
- [51] Xiao Hua, Valera-Medina Agustín, Bowen Philip J. Modeling combustion of ammonia/hydrogen fuel blends under gas turbine conditions. *Energy Fuels* 2017;31(8):8631–42.
- [52] Kumar Praveen, Meyer Terrence R. Experimental and modeling study of chemical-kinetics mechanisms for H₂-NH₃-air mixtures in laminar premixed jet flames. *Fuel* 2013;108:166–76.
- [53] Xiao Hua, Valera-Medina Agustín. Chemical kinetic mechanism study on pre-mixed combustion of ammonia/hydrogen fuels for gas turbine use. *J Eng Gas Turb Power* 2017;139(8).
- [54] Mashruk Syed, Kovaleva Marina, Tung Chong Cheng, Hayakawa Akihiro, Okafor Ekenechukwu C, Valera-Medina Agustín. Nitrogen oxides as a by-product of ammonia/hydrogen combustion regimes. *Chem Eng Trans* 2021;89:613–8.
- [55] Mashruk Syed, Kovaleva Marina, Alnasif Ali, Chong Cheng Tung, Hayakawa Akihiro, Okafor Ekenechukwu C, Valera-Medina Agustín. Nitrogen oxide emissions analysis in ammonia/hydrogen/air premixed swirling flames. *Energy* 2022;260:125183.
- [56] Duynslaegher Catherine, Jeanmart Hervé, Vandooren Jacques. Flame structure studies of premixed ammonia/hydrogen/oxygen/argon flames: experimental and numerical investigation. *Proc Combust Inst* 2009;32(1):1277–84.
- [57] Goldmeier Jeffrey. 16 near-zero-and zero-carbon fuels in industrial gas turbines. 2023. [Online; accessed 10-Aug-2023]. <https://www.ge.com/news/press-releases/ge-and-ihl-sign-memorandum-of-understanding-to-develop-gas-turbines-that-can-operate>.
- [58] 2023. [Online; accessed 10-Aug-2023]. <https://www.ge.com/news/press-releases/ge-and-ihl-sign-memorandum-of-understanding-to-develop-gas-turbines-that-can-operate>.
- [59] Valera-Medina Agustín, Gutesa M, Xiao H, Pugh D, Giles A, Goktepe B, Marsh R, Bowen P. Premixed ammonia/hydrogen swirl combustion under rich fuel conditions for gas turbines operation. *Int J Hydrogen Energy* 2019;44(16):8615–26.
- [60] Aalrebai Odi Fawwaz, Al Assaf Anwar Hamdan, Amhamed Abdulkarem, Swaminathan Nedunchezian, Hewlett Sally. Ammonia-hydrogen-air gas turbine cycle and control analyses. *Int J Hydrogen Energy* 2022;47(13):8603–20.
- [61] Bitter Rick, Mohiuddin Taqi, Nawrocki Matt. LabVIEW: Advanced programming techniques. CRC Press; 2017.
- [62] Morley Chris. Gaseq: a chemical equilibrium program for windows. 2005, Ver. 0.79.
- [63] Božo M Guteša, Vigueras-Zuniga MO, Buffi Marco, Seljak Tine, Valera-Medina Agustín. Fuel rich ammonia-hydrogen injection for humidified gas turbines. *Appl Energy* 2019;251:113334.
- [64] Božo M Guteša, Mashruk Syed, Zitouni S, Valera-Medina Agustín. Humidified ammonia/hydrogen RQL combustion in a trigeneration gas turbine cycle. *Energy Convers Manage* 2021;227:113625.
- [65] Reuß Markus, Grube Thomas, Robinus Martin, Stolten Detlef. A hydrogen supply chain with spatial resolution: Comparative analysis of infrastructure technologies in Germany. *Appl Energy* 2019;247:438–53.
- [66] Li Miao, Ming Pingwen, Huo Ran, Mu Hailin, Zhang Cunman. Optimizing design and performance assessment of a sustainability hydrogen supply chain network: A multi-period model for China. *Sustainable Cities Soc* 2023;92:104444.
- [67] Kim Changjong, Cho Sang Hoon, Cho Sung Min, Na Youngseung, Kim Sangwon, Kim Dong Kyu. Review of hydrogen infrastructure: the current status and roll-out strategy. *Int J Hydrogen Energy* 2023;48(5):1701–16.
- [68] Ren Xusheng, Dong Lichun, Xu Di, Hu Baoshan. Challenges towards hydrogen economy in China. *Int J Hydrogen Energy* 2020;45(59):34326–45.
- [69] Sdanghi Giuseppe, Maranzana Gaël, Celzard Alain, Fierro Vanessa. Review of the current technologies and performances of hydrogen compression for stationary and automotive applications. *Renew Sustain Energy Rev* 2019;102:150–70.
- [70] Gardiner Monterey. Energy requirements for hydrogen gas compression and liquefaction as related to vehicle storage needs. In: *DOE hydrogen and fuel cells program record*, Vol. 9013. DOE Washington, DC, USA; 2009.
- [71] Yanxing Zhao, Maoqiong Gong, Yuan Zhou, Xueqiang Dong, Jun Shen. Thermodynamics analysis of hydrogen storage based on compressed gaseous hydrogen, liquid hydrogen and cryo-compressed hydrogen. *Int J Hydrogen Energy* 2019;44(31):16833–40.

- [72] Sapountzi Foteini M, Gracia Jose M, Fredriksson Hans OA, Niemantsverdriet JW Hans, et al. Electrocatalysts for the generation of hydrogen, oxygen and synthesis gas. *Prog Energy Combust Sci* 2017;58:1–35.
- [73] Chi Jun, Yu Hongmei. Water electrolysis based on renewable energy for hydrogen production. *Chin J Catal* 2018;39(3):390–4.
- [74] Mikulčić Hrvoje, Baleta Jakov, Wang Xuebin, Wang Jin, Qi Fengsheng, Wang Fan. Numerical simulation of ammonia/methane/air combustion using reduced chemical kinetics models. *Int J Hydrogen Energy* 2021;46(45):23548–63.
- [75] Wang Jing, Jiang Xi Zhuo, Luo Kai H. Exploring reaction mechanism for ammonia/methane combustion via reactive molecular dynamics simulations. *Fuel* 2023;331:125806.
- [76] Tang Guang, Jin Pengfei, Bao Yulei, Chai Wai Siong, Zhou Lei. Experimental investigation of premixed combustion limits of hydrogen and methane additives in ammonia. *Int J Hydrogen Energy* 2021;46(39):20765–76.
- [77] Kurata Osamu, Iki Norihiko, Matsunuma Takayuki, Inoue Takahiro, Tsujimura Taku, Furutani Hirohide, Kobayashi Hideaki, Hayakawa Akihiro. Performances and emission characteristics of NH₃-air and NH₃CH₄-air combustion gas-turbine power generations. *Proc Combust Inst* 2017;36(3):3351–9.
- [78] Iki Norihiko, Kurata Osamu, Matsunuma Takayuki, Inoue Takahiro, Tsujimura Taku, Furutani Hirohide, Kobayashi Hideaki, Hayakawa Akihiro. Operation and flame observation of micro gas turbine firing ammonia. In: *Turbo expo: power for land, sea, and air*, Vol. 50954. American Society of Mechanical Engineers; 2017, V008T26A018.
- [79] Valera-Medina Agustin, Marsh Richard, Runyon Jon, Pugh Daniel, Beasley Paul, Hughes Timothy, Bowen Phil. Ammonia-methane combustion in tangential swirl burners for gas turbine power generation. *Appl Energy* 2017;185:1362–71.
- [80] Xiao Hua, Valera-Medina Agustin, Marsh Richard, Bowen Philip J. Numerical study assessing various ammonia/methane reaction models for use under gas turbine conditions. *Fuel* 2017;196:344–51.
- [81] Ávila Cristian D, Cardona Santiago, Abdullah Marwan, Younes Mourad, Jamal Aqil, Guiberti Thibault F, Roberts William L. Experimental assessment of the performance of a commercial micro gas turbine fueled by ammonia-methane blends. *Appl Energy Combust Sci* 2023;13:100104.
- [82] Ilbas Mustafa, Kumuk Osman, Alabas Bugrahan. Experimental investigation of the combustion instability behaviors and NO_x emissions in an ammonia/methane fueled low eddy premixed burner. *J Energy Inst* 2023;108:101241.
- [83] Wang Shixing, Wang Zhihua, Elbaz Ayman M, Han Xinlu, He Yong, Costa Mário, Konnov Alexander A, Roberts William L. Experimental study and kinetic analysis of the laminar burning velocity of NH₃/syngas/air, NH₃/CO/air and NH₃/H₂/air premixed flames at elevated pressures. *Combust Flame* 2020;221:270–87.
- [84] Mousavi S Mahmood, Lee Bok Jik, Kim Jinyoung, Sotoudeh Freshteh, Chun Byoungjoo, Jun Daeyoung, Karimi Nader, Esfahani Javad Abolfazli. On the effects of adding syngas to an ammonia-MILD combustion regime—A computational study of the reaction zone structure. *Int J Hydrogen Energy* 2023.
- [85] Kang Lianwei, Pan Weiguo, Zhang Jiakai, Wang Wenhuan, Tang Congwei. A review on ammonia blends combustion for industrial applications. *Fuel* 2023;332:126150.
- [86] Chen Zhiqiang, Jiang Yong. Numerical investigation of the effects of H₂/CO/syngas additions on laminar premixed combustion characteristics of NH₃/air flame. *Int J Hydrogen Energy* 2021;46(21):12016–30.
- [87] Cai Ziheng, Huang Mingming, Wei Gaofeng, Liu Zhenxian, Fang Huiwei, Song Yang, Zhang Mengting, Wang Weiming, Ming Zhaoqing, Wang Jihang, et al. Numerical study of the effect of pressure on the combustion characteristics of ammonia/coal-derived syngas mixture under gas turbine operating conditions. *Fuel* 2023;347:128463.
- [88] Han Xinlu, Wang Zhihua, Costa Mario, Sun Zhiwei, He Yong, Cen Kefa. Experimental and kinetic modeling study of laminar burning velocities of NH₃/air, NH₃/H₂/air, NH₃/CO/air and NH₃/CH₄/air premixed flames. *Combust Flame* 2019;206:214–26.
- [89] Mei Bowen, Ma Siyuan, Zhang Yan, Zhang Xiaoyuan, Li Wei, Li Yuyang. Exploration on laminar flame propagation of ammonia and syngas mixtures up to 10 atm. *Combust Flame* 2020;220:368–77.
- [90] Bicer Yusuf, Khalid Farrukh. Life cycle environmental impact comparison of solid oxide fuel cells fueled by natural gas, hydrogen, ammonia and methanol for combined heat and power generation. *Int J Hydrogen Energy* 2020;45(5):3670–85.
- [91] Strube R, Pellegrini G, Manfrida G. The environmental impact of post-combustion CO₂ capture with MEA, with aqueous ammonia, and with an aqueous ammonia-ethanol mixture for a coal-fired power plant. *Energy* 2011;36(6):3763–70.
- [92] Karabeyoglu Arif, Evans Brian, Stevens Jose, Cantwell Brian, Micheletti Dave. Development of ammonia based fuels for environmentally friendly power generation. In: *10th international energy conversion engineering conference*. 2012, p. 4055.
- [93] Xu Xiaowei, Liu Enlong, Zhu Neng, Liu Fanfu, Qian Feng. Review of the current status of ammonia-blended hydrogen fuel engine development. *Energies* 2022;15(3):1023.
- [94] Tan Eric CD, Talmadge Michael, Dutta Abhijit, Hensley Jesse, Schaidle Josh, Bidy Mary, Humbird David, Snowden-Swan Lesley J, Ross Jeff, Sexton Danielle, et al. Process design and economics for the conversion of lignocellulosic biomass to hydrocarbons via indirect liquefaction. Thermochemical research pathway to high-octane gasoline Blendstock through methanol/dimethyl ether intermediates. Technical Report, Golden, CO (United States): National Renewable Energy Lab. (NREL); 2019.
- [95] Yuan Jiahai, Na Chunling, Lei Qi, Xiong Minpeng, Guo Jingsheng, Hu Zheng. Coal use for power generation in China. *Resour Conserv Recy* 2018;129:443–53.
- [96] Clark Richard, Zucker Noah, Urpelainen Johannes. The future of coal-fired power generation in southeast Asia. *Renew Sustain Energy Rev* 2020;121:109650.
- [97] Zhao Changhong, Zhang Weirong, Wang Yang, Liu Qilin, Guo Jingsheng, Xiong Minpeng, Yuan Jiahai. The economics of coal power generation in China. *Energy Policy* 2017;105:1–9.
- [98] Shearer Christine, Fofrich Robert, Davis Steven J. Future CO₂ emissions and electricity generation from proposed coal-fired power plants in India. *Earth's Future* 2017;5(4):408–16.
- [99] Oh Tick Hui. Carbon capture and storage potential in coal-fired plant in Malaysia—A review. *Renew Sustain Energy Rev* 2010;14(9):2697–709.
- [100] Khayyam Umer, Nazar Umair, et al. Energy production and CO₂ emissions: The case of coal fired power plants under China Pakistan economic corridor. *J Clean Prod* 2021;281:124974.
- [101] Descamps C, Bouallou Chakib, Kanniche Mohamed. Efficiency of an integrated gasification combined cycle (IGCC) power plant including CO₂ removal. *Energy* 2008;33(6):874–81.
- [102] Christou Costas, Hadjipaschalis Ioannis, Poullikkas Andreas. Assessment of integrated gasification combined cycle technology competitiveness. *Renew Sustain Energy Rev* 2008;12(9):2459–71.
- [103] Ren Siyue, Feng Xiao, Wang Yufei. Energy evaluation of the integrated gasification combined cycle power generation systems with a carbon capture system. *Renew Sustain Energy Rev* 2021;147:111208.
- [104] Brown T. Chugoku electric completes successful trial, seeks patent for ammonia co-firing technology. *Ammonia Energy* 2017.
- [105] Pashchenko Dmitry. Industrial furnaces with thermochemical waste-heat recuperation by coal gasification. *Energy* 2021;221:119864.
- [106] Wan Zhijian, Tao Youkun, Shao Jing, Zhang Yinghui, You Hengzhi. Ammonia as an effective hydrogen carrier and a clean fuel for solid oxide fuel cells. *Energy Convers Manage* 2021;228:113729.
- [107] Makepeace Joshua W, He Teng, Weidenthaler Claudia, Jensen Torben R, Chang Fei, Vegge Tejs, Ngene Peter, Kojima Yoshitsugu, de Jongh Petra E, Chen Ping, et al. Reversible ammonia-based and liquid organic hydrogen carriers for high-density hydrogen storage: Recent progress. *Int J Hydrog Energy* 2019;44(15):7746–67.
- [108] Aziz Muhammad, Oda Takuya, Kashiwagi Takao. Comparison of liquid hydrogen, methylcyclohexane and ammonia on energy efficiency and economy. *Energy Procedia* 2019;158:4086–91.
- [109] Lamb Krystina E, Dolan Michael D, Kennedy Danielle F. Ammonia for hydrogen storage; A review of catalytic ammonia decomposition and hydrogen separation and purification. *Int J Hydrogen Energy* 2019;44(7):3580–93.
- [110] Morlanés Natalia, Katikaneni Sai P, Paglieri Stephen N, Harale Aadesh, Solami Bandar, Sarathy S Mani, Gascon Jorge. A technological roadmap to the ammonia energy economy: Current state and missing technologies. *Chem Eng J* 2021;408:127310.
- [111] Hu Yinghui, Tao Bowen, Shang Fei, Zhou Meixue, Hao Dongyu, Fan Ruiqing, Xia Debin, Yang Yulin, Pang Aimin, Lin Kaifeng. Thermal decomposition of ammonium perchlorate over perovskite catalysts: Catalytic decomposition behavior, mechanism and application. *Appl Surf Sci* 2020;513:145849.
- [112] Chehade Ghassan, Dincer Ibrahim. Progress in green ammonia production as potential carbon-free fuel. *Fuel* 2021;299:120845.
- [113] Badakhsh Arash, Kwak Yeonsu, Lee Yu-Jin, Jeong Hyangsoo, Kim Yongmin, Sohn Hyuntae, Nam Suk Woo, Yoon Chang Won, Park Chan Woo, Jo Young Suk. A compact catalytic foam reactor for decomposition of ammonia by the joule-heating mechanism. *Chem Eng J* 2021;426:130802.
- [114] Badakhsh Arash, Cha Junyoung, Park Yongha, Lee Yu-Jin, Jeong Hyangsoo, Kim Yongmin, Sohn Hyuntae, Nam Suk Woo, Yoon Chang Won, Park Chan Woo, et al. Autothermal recirculating reactor (ARR) with Cu-BN composite as a stable reactor material for sustainable hydrogen release from ammonia. *J Power Sources* 2021;506:230081.
- [115] Pashchenko Dmitry, Mustafin Ravil. Ammonia decomposition in the thermochemical waste-heat recuperation systems: a view from low and high heating value. *Energy Convers Manage* 2022;251:114959.
- [116] Wang Bingzheng, Kong Hui, Wang Hongsheng, Wang Yipu, Hu Xuejiao. Kinetic and thermodynamic analyses of mid/low-temperature ammonia decomposition in solar-driven hydrogen permeation membrane reactor. *Int J Hydrogen Energy* 2019;44(49):26874–87.
- [117] Badescu Viorel. Optimal design and operation of ammonia decomposition reactors. *Int J Energy Res* 2020;44(7):5360–84.
- [118] Popov SK, Svistunov IN, Garyaev AB, Serikov EA, Temyrkanova EK. The use of thermochemical recuperation in an industrial plant. *Energy* 2017;127:44–51.

- [119] Pashchenko Dmitry. Energy optimization analysis of a thermochemical exhaust gas recuperation system of a gas turbine unit. *Energy Convers Manage* 2018;171:917–24.
- [120] Tartakovsky L, Sheintuch M. Fuel reforming in internal combustion engines. *Prog Energy Combust Sci* 2018;67:88–114.
- [121] Poran A, Thawko A, Eyal A, Tartakovsky L. Direct injection internal combustion engine with high-pressure thermochemical recuperation—experimental study of the first prototype. *Int J Hydrogen Energy* 2018;43(27):11969–80.
- [122] Poran Arnon, Tartakovsky Leonid. Energy efficiency of a direct-injection internal combustion engine with high-pressure methanol steam reforming. *Energy* 2015;88:506–14.
- [123] Pashchenko Dmitry. Low-grade heat utilization in the methanol-fired gas turbines through a thermochemical fuel transformation. *Therm Sci Eng Prog* 2022;36:101537.
- [124] Pashchenko Dmitry. First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming. *Energy* 2018;143:478–87.
- [125] Gaber Christian, Demuth Martin, Schluckner Christoph, Hochenauer Christoph. Thermochemical analysis and experimental investigation of a recuperative waste heat recovery system for the tri-reforming of light oil. *Energy Convers Manage* 2019;195:302–12.
- [126] Chakrabarti Reetam, Colby Joshua L, Schmidt Lanny D. Effects of biomass inorganics on rhodium catalysts: I. Steam methane reforming. *Appl Catal B* 2011;107(1):88–94.
- [127] Sikirica Steve, Kurek H, Kozlov A, Khinkis M. Thermochemical recuperation improves furnace thermal efficiency. *Heat Treat Progress* 2007;7(5):28.
- [128] Su Bosheng, Huang Yupeng, Wang Yilin, Huang Zhi, Yuan Shuo, Huang Qiteng, Xu Zhilong, Lin Feng. Novel ammonia-driven chemically recuperated gas turbine cycle based on dual fuel mode. *Appl Energy* 2023;343:121184.
- [129] Bartolome Pedro Santos, Van Gerven Tom. A comparative study on Aspen hysys interconnection methodologies. *Comput Chem Eng* 2022;162:107785.
- [130] Kartal Furkan, Sezer Senem, Özveren Uğur. Investigation of steam and CO₂ gasification for biochar using a circulating fluidized bed gasifier model in Aspen HYSYS. *J CO₂ Util* 2022;62:102078.
- [131] Larson AT, Dodge RL. The ammonia equilibrium. *J Am Chem Soc* 1923;45(12):2918–30.
- [132] Peters Stefan, Abdel-Mageed Ali M, Wohlrab Sebastian. Thermocatalytic ammonia decomposition—status and current research demands for a carbon-free hydrogen fuel technology. *ChemCatChem* 2023;15(2):e202201185.
- [133] Ao Rui, Lu Ruihua, Leng Guanghui, Zhu Youran, Yan Fuwu, Yu Qinghua. A review on numerical simulation of hydrogen production from ammonia decomposition. *Energies* 2023;16(2):921.
- [134] Cechetto Valentina, Di Felice Luca, Martinez Rocio Gutierrez, Plazaola Alba Aratibel, Gallucci Fausto. Ultra-pure hydrogen production via ammonia decomposition in a catalytic membrane reactor. *Int J Hydrog Energy* 2022;47(49):21220–30.
- [135] Alva Guruprasad, Liu Lingkun, Huang Xiang, Fang Guiyin. Thermal energy storage materials and systems for solar energy applications. *Renew Sustain Energy Rev* 2017;68:693–706.
- [136] Mellor Alexander, Alvarez D Alonso, Guarracino Ilaria, Ramos Alba, Lacasta A Riverola, Llin L Ferre, Murrell AJ, Paul DJ, Chemisana D, Markides CN, et al. Roadmap for the next-generation of hybrid photovoltaic-thermal solar energy collectors. *Sol Energy* 2018;174:386–98.
- [137] Wang Zhou-jun, Song Hui, Liu Huimin, Ye Jinhua. Coupling of solar energy and thermal energy for carbon dioxide reduction: status and prospects. *Angew Chem Int Ed* 2020;59(21):8016–35.
- [138] Qiu Yu, Xu Yucong, Li Qing, Wang Jikang, Wang Qiliang, Liu Bin. Efficiency enhancement of a solar trough collector by combining solar and hot mirrors. *Appl Energy* 2021;299:117290.
- [139] Saini Manish, Sharma Abhishek, Singh Varun Pratap, Dwivedi Gaurav, Jain Sidharth. Solar thermal receivers—a review. In: *Advancement in materials, manufacturing and energy engineering*, Vol. II: Select proceedings of ICAMME 2021. Springer; 2022, p. 311–25.
- [140] Gómez-Hernández Jesús, González-Gómez PA, Briongos JV, Santana D. Technical feasibility analysis of a linear particle solar receiver. *Sol Energy* 2020;195:102–13.